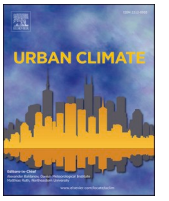
Urban Climate 55 (2024) 101982



Contents lists available at ScienceDirect

Urban Climate

journal homepage: www.elsevier.com/locate/uclim





Towards an improved representation of the urban heat island effect : A multi-scale application of XGBoost for madrid

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ARTICLE INFO

Keywords:
Surface urban heat island
Machine learning
Extreme gradient boosting
Global climate modelling
City-scale climate
Land-use

ABSTRACT

Cities are considered local "hotspots" of climate change, therefore, the improvement of urban present climate description as well as future projections is paramount for designing adaptation and mitigation strategies. Physically-based numerical models often have coarse resolutions and do not present parametrisations to adequately represent physical processes at the urban scale. This article presents an innovative application of XGBoost (a machine learning approach) as an alternative to explore and improve the description of the climate of Madrid. XGBoost's ability to reproduce the present climate of 2-m air temperature and land surface temperature (LST), and the urban heat island (UHI) effect, was assessed. XGBoost was trained with a set of ERA5 predictors (0.25°) and calibrated with observations from ground stations (2000−2022) and remote sensing data (2004–2022). Several sensitivity cases were performed to assess the results dependency to predictors and their resolution. XGBoost was evaluated at daily scale for the 2-m maximum and minimum temperatures (Tmax and Tmin, respectively) and LST, and at hourly scale for LST. Overall, XGBoost reveals a good performance and significant added value against ERA5 for all variables considered and for both UHI and surface UHI. This study presents XGBoost as a promising technology to describe urban climate.

1. Introduction

Cities are considered local "hotspots" of climate change impacts (IPCC, 2021). Despite urban areas constituting a small fraction of the total land surface (3% for urban areas, 0.65% for built-up areas, and 0.45% for impervious surface areas, excluding Antarctica and Greenland; Liu et al., 2014), they accommodate approximately 55% of the global population - a fraction expected to increase to 60% and 68% by 2030 and 2050, respectively (United Nations et al., 2018). Moreover, cities are responsible for 75% of global energy-related greenhouse gas emissions resulting from concentrations in infrastructure and economic activities (IPCC, 2013). Therefore, it is crucial to understand and simulate the evolution of the urban climate to design adaptation and mitigation strategies to climate change.

The extensive surface modification resulting from the urbanisation process plays a crucial role in driving local atmospheric variables, thereby giving rise to an urban climate (Masson et al., 2020). The strong changes in land-use of urban areas include modifications of the surface's physical properties and its morphology. Regarding the former, the typical dry and impervious surface materials

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https://doi.org/10.1016/j.uclim.2024.101982
Received 21 December 2023; Received in revised form 28 April 2024; Accepted 26 May 2024
Available online 14 June 2024
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prevailing in cities are characterised by distinct radiative, thermal, and moisture properties leading to higher surface temperatures. Concerning the morphology, the cities' structure is responsible for increased trapping and absorption of solar radiation, which arises from the interception of surface reflected radiation, reducing the overall reflected radiation and decreasing the city albedo (Oke et al., 2017). As a consequence, the urban areas experience reduced evapotranspiration and the decreased nighttime cooling in built-up areas (Masson et al., 2020). Therefore, cities have typically warmer air and surface temperatures compared to surrounding rural areas.. This well-known phenomenon is designated Urban Heat Island (UHI), usually is stronger at nighttime and enhances during heatwaves (Kong et al., 2021; Shu et al., 2023). A similar effect, the Surface Urban Heat Island (SUHI), describes the difference between urban and rural Land Surface Temperatures (LSTs). The establishment and further development of these effects are enhanced under weak regional wind flow that attenuate thermal advection (Oke et al., 2017).

Although relatively straightforward to conceptualise and determine, the UHI phenomenon varies across cities due to multiple factors related to urban internal structure (e.g., size and density of buildings) and the surrounding environment (e.g., geography and background climate; Oke et al., 2017). In complex geographical areas, such as coastal, mountainous, and cold winter climates, the UHI effect assessment and its development becomes more intricate (Masson et al., 2020). Moreover, the additional heat accumulated in cities leads to a plethora of social, economic, and health impacts that are challenging to predict. The UHI effect has been demonstrated to intensify locally heat waves, thereby increasing exposure to heat stress, and elevating the mortality rate in cities compared to the surrounding rural areas (Tan et al., 2010). Within the context of climate change, the enhanced heat stress inherent to urban environments will expose inhabitants to more frequent and severe extreme events, including temperature extremes, heatwaves, and droughts (IPCC, 2021). As an example, heatwaves are expected to enhance in frequency, intensity, and duration (Vautard et al., 2013; Perkins, 2015; Cardoso et al., 2023), exposing cities even more to this phenomenon. The evolution of the UHI effect under climate change conditions has a high uncertainty, showing contrasting results using different methods and for different cities. Nevertheless, when compared to the projected future climate warming in urban and rural areas, the UHI change is lower by one order of magnitude, with a high degree of confidence. However, the ongoing urbanisation process along with the aggravation of extreme weather events will make urban citizens more exposed to global warming effects (IPCC, 2021). Moreover, the number of local observations in cities is still rather limited in most of the cities to fully understand its climate heterogeneities and related impacts. Thus, it is crucial to describe the urban climate in present climate in a detailed manner to understand the present and future impacts on the urban environments and society.

Numerical models play a crucial role in climate studies and numerical weather prediction. The state-of-the-art Earth System and Global Climate Models (ESGCMs) have a horizontal spatial resolution of the order of 100 km, which is considered too coarse to accurately capture regional to local scale processes (Randall et al., 2007; Soares et al., 2012; Rummukainen, 2016), including surface heterogeneities associated with urban city patterns (Oleson et al., 2011; Garuma, 2018; Langendijk et al., 2019). To address this limitation, downscaling methods are employed to provide a better representation of local to regional processes associated with surface heterogeneities. Downscaling approaches (Le Roux et al., 2018) are broadly categorised into Dynamical Downscaling Methods and Statistical Downscaling Methods (SDMs). Dynamical Downscaling Methods employ Regional Climate Models (RCMs), that are forced by ESGCMs at the boundaries, and dynamically transfer large-scale features to the limited area domain (Christensen et al., 2007; Dickinson et al., 1989; Giorgi and Bates, 1989). Typically, these models have a grid resolution of the order of 10 km, substantially improving the description of local to regional atmospheric processes and feedback mechanisms when compared to ESGCMs (Rummukainen, 2016; Soares et al., 2012; Soares et al., 2017a, 2017b; Cardoso et al., 2019). However, in the scope of urban climate studies, this resolution is still limited for representing many aspects of the local climate. Recent efforts within the scientific community aim to build RCMs with higher spatial resolutions (2 to 3 km), named often convective permitting or km-scale simulations, allowing a more detailed description of fundamental local climate processes, including convective and urban processes (Prein et al., 2015; Coppola et al., 2020; Ban et al., 2021; Pichelli et al., 2021; Shu et al., 2023; Soares et al., 2022). However, this approach comes with the disadvantage of high computational cost, often limiting the ability to simulate many cities over a wide climatic temporal span. In contrast, SDMs are a computationally less expensive approach. These methods rely on establishing empirical relationships between large-scale atmospheric climate patterns (predictors) and local climate observations (predictands). As SDMs use observations for training, they do not show the systematic biases often exhibited by RCMs. However, their architectures lack representation of physical processes (Fowler et al., 2007; Nikulin et al., 2018; Soares et al., 2019; Wilby et al., 1998).

One particularly promising and continuously advancing branch of SDMs is based on Machine Learning (ML) algorithms. ML-based downscaling is a data-driven method in contrast to the incorporation of physical laws in numerical simulations. These are divided into supervised and unsupervised learning (Alloghani et al., 2020). In supervised learning, the algorithm learns the connection between pairs of inputs and outputs. In unsupervised learning, where no specific output is given, the goal is to identify structures or connections within the data. There is presently a widely accepted acknowledgment of the potential advantages of developing artificial intelligence (AI) methods, as machine learning, for climate and weather applications, particularly for better local climate characterisation (Reichstein et al., 2019; Bonavita et al., 2021; Zheng et al., 2021).

In the current study, a ML method, Extreme Gradient Boosting (XGBoost), is explored to describe several relevant urban climate features. XGBoost is a supervised ML algorithm based on gradient-boosted regression trees (Chen and Guestrin, 2016). It captures non-linear relationships between a set of predictors and observations, generating predictions for a set of predictands. XGBoost is widely applied in many data science fields (Chen and Guestrin, 2016); however, its application in Earth Sciences, specifically in urban climate, remains relatively limited. An example of XGBoost's application in urban climate studies was proposed by Zheng et al. (2021). The authors applied XGBoost in a climate emulator framework to assess the inter-model variability in projections of local urban heat waves globally under a future high-emission scenario. Another study, Fan et al. (2018), compared the accuracy, stability, and computational costs of XGBoost with the Support Vector Machine model (another ML method) for modelling global solar radiation in humid regions

using limited temperature and precipitation data.

The Iberian Peninsula is a recognised climate change "hotspot" (Planton et al., 2012; Russo et al., 2019; Cos et al., 2022) due to its vulnerability to warming and drying conditions (Cardoso et al., 2019; Soares et al., 2017a; Soares et al., 2017b; Soares and Lima, 2022), leading to significant impacts on extreme events such as heatwaves, wildfires and droughts, (Cardoso et al., 2019; Bento et al., 2022, 2023; Soares et al., 2023; Soares et al., 2024). In this study, XGBoost potential is assessed for Madrid, the largest Iberian city, with a total population of 3.3 million (6.9 million considering the metropolitan area). Urban climate studies focused on Madrid generally estimate strong positive UHI and SUHI effects during the nighttime and weaker, sometimes slightly negative intensities during daytime (Dialesandro et al., 2019; Fernández García et al., 2016; Núñez Peiró et al., 2017; Núñez-Peiró et al., 2021a; Rasilla et al., 2019a; Sobrino et al., 2013a; Zheng and Roca, 2021).

The main goal of this study is to evaluate the ability of the XGBoost in depicting the present climate of Madrid by downscaling ERA5 to the local city scale. Several XGBoost setups were compared to understand the sensitivity of the method to the selection of predictors (different surface-level and height/pressure-level variables). Additionally, two different predictors resolution is also assessed. Subsequently, the overall better setup, concerning accuracy and model data availability, is applied to characterise the present urban climate variables (maximum and minimum 2-m temperatures (T_{max} and T_{min}), LST, UHI and SUHI) in Madrid and compared with observations. T_{max}, T_{min}, and UHI are evaluated at the monthly and daily scales. LST and the SUHI effect are evaluated at the monthly, daily, and hourly scales. This assessment allows to understand the potential applicability of XGBoost, forced by different climate output products, from ESGCM to RCMs, available at very diverse spatial and temporal scales.

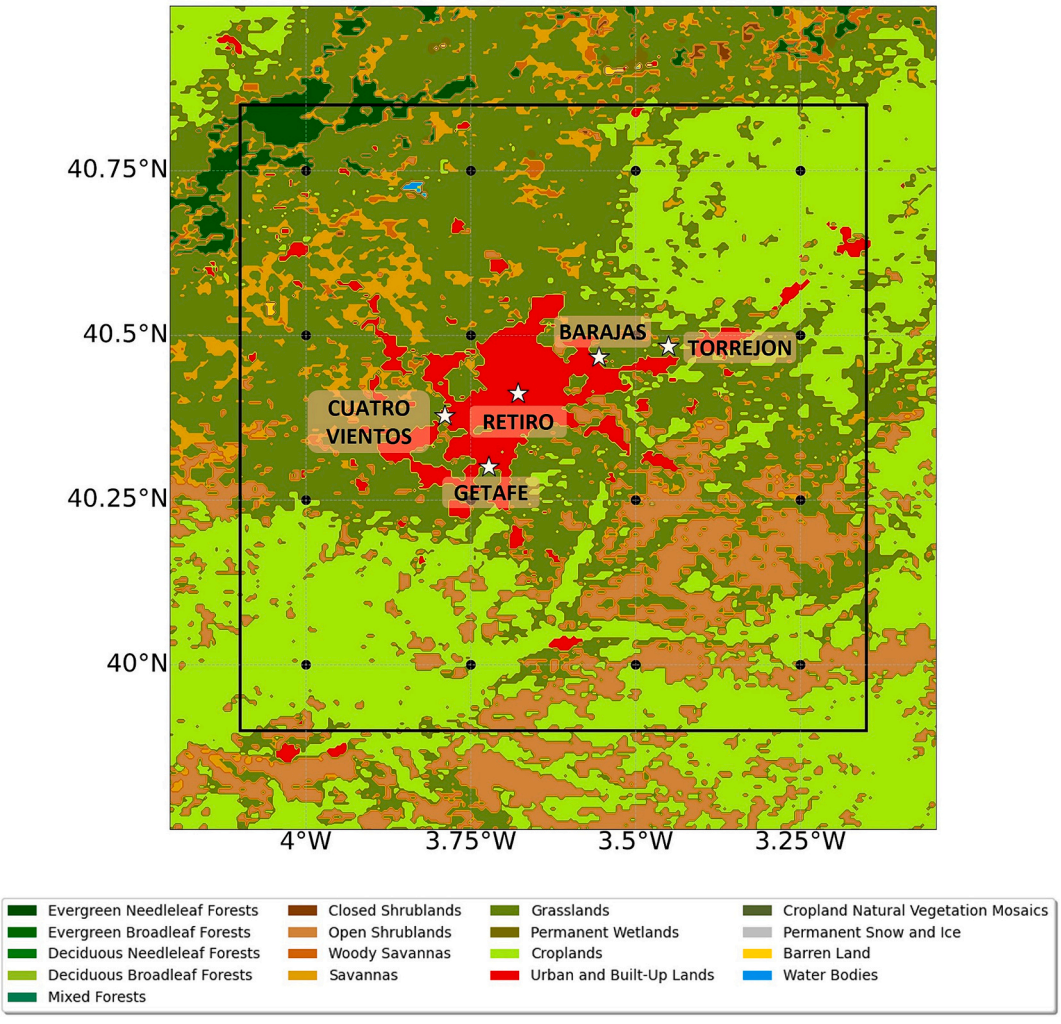


Fig. 1. Land Cover Type (product from Moderate-Resolution Imaging Spectroradiometer (MODIS) at 500 m resolution for 2015) over the study domain delimited by the black square. The white stars represent the urban (Retiro) and rural (Torrejon, Barajas, Cuatro Vientos and Getafe) meteorological stations considered in this study. The black circles represent the ERA5 grid points.

2. Data and methods

2.1. Study area

The Iberian Peninsula has a highly heterogeneous (Cardoso et al., 2013; Gómez-Navarro et al., 2011; Soares et al., 2023; Soares et al., 2024). Whereas the southern half of Iberia is characterised by a Mediterranean climate with dry and warm summers, the northern half is dominated by a Mediterranean oceanic climate with milder summers, mountainous areas with oceanic climate and SE Spain and Portugal with a semi-arid climate (Kottek et al., 2006; Cardoso et al., 2013). The focus of the present study is the city of Madrid, located in the centre of Spain and transitioning from a Mediterranean to a cold semi-arid climate (*Csa* and *BSk*, respectively, according to the Köppen Climate classification; Kottek et al., 2006; Beck et al., 2018). Moreover, considering the ASHRAE classification, the Barajas and Torrejon stations have Warm-dry (3B) and Mixed-dry (4B) climates types, respectively (ASHRAE, 2021). Madrid is slightly influenced by the Central System mountain range, situated approximately 50 km north of the city. The city's climate is characterised by hot and dry summers and cold winters, with sparse precipitation that occurs mostly in autumn and spring (Sobrino et al., 2013b), and cloudless days predominantly during summer (Núñez-Peiró et al., 2021b). Located in a relatively flat area, Madrid reaches a maximum elevation of 700 m above sea level in the northern city limit and a minimum elevation of 550 m along the Manzanares River (Fernández García et al., 1996). The nearby mountain range, which divides the Meseta Central plateau, influences the wind direction (mostly NE–SW) and creates a temperature gradient from cooler temperatures in the north to warmer temperatures in the south. The Manzanares River, flowing from NW to SE, creates a topographic depression and channels cold air from the mountains into the city, further contributing to the local temperature gradient.

The study domain for Madrid is delimited by the black square (4.09°E – 3.15°E; 39.9°N – 40.85°N) in Fig. 1.

2.2. Local ground stations

To characterise the urban climate of Madrid, observational data from five meteorological stations in the city and their surroundings were used (Fig. 1; see Table 1 for a summary), including daily maximum and minimum 2-m air temperatures (T_{max} and T_{min}, respectively), spanning the period from 2000 to 2022. The observations were obtained from the European Climate Assessment & Dataset (ECA&D), which incorporates a quality control check. The Local Climate Zones (LCZ) classification is a relevant tool for characterising the surrounding environment of the meteorological stations used in this study. The combination of built and land cover types provided by the LCZ classification is essential to understand the urbanisation level of the sites, allowing to differentiate between urban and rural stations (Oke et al., 2017; Aslam and Rana, 2022; Bechtel et al., 2019; Chen et al., 2023; Kotharkar and Bagade, 2018; Perera and Emmanuel, 2018; Richard et al., 2018; Shi et al., 2018; Stewart and Oke, 2012; Yang et al., 2020; Zhao et al., 2020; Zheng et al., 2023). Predominant LCZs in Madrid include compact midrise (LCZ 2) and open midrise (LCZ 5), with the low–rise class (LCZ 8) dominating the south-western periphery (Rasilla et al., 2019).

The Retiro station is located in the centre of Madrid at El Retiro urban park (LCZ B), approximately 100 m from a dense urban area (LCZ 2). Despite being surrounded by a vegetated area, Retiro is commonly used as an urban reference in studies that characterise Madrid's urban temperature due to its proximity to dense urbanisation (Michel et al., 2021; Núñez-Peiró et al., 2021; Rasilla et al., 2019; Salamanca et al., 2012). The other stations are installed in airports, over low plant areas (LCZ D), and nearby urban areas with varying urban densities (Rasilla et al., 2019). The selected set of meteorological stations was previously studied by Rasilla et al. (2019) to characterise Madrid's UHI effect during heatwaves. With this set of stations (Retiro, Torrejon, Barajas, Cuatro Vientos, and Getafe) Rasilla et al. (2019) achieved results which are agreement with other state-of-the art studies of UHI and SUHI description for Madrid (Dialesandro et al., 2019; Fernández García et al., 2016; Núñez Peiró et al., 2017; Núñez-Peiró et al., 2021; Sobrino et al., 2013), depicting a clear strong positive/weak negative signal at nighttime/daytime of UHI. Noteworthy, the lack of other stations near Madrid with long-term high quality daily observational series of temperature limited this selection. Therefore, the Torrejon, Barajas, Cuatro Vientos, and Getafe, which are located relatively far from the city centre and at the periphery of the urban area of Madrid were used in this study to describe an urban-periphery temperature, thus allowing to estimate the UHI effect. Hereafter, although acknowledging the approximation character, the latter stations are classified as "rural" to simplify the analysis.

2.3. Land cover type

The Land Cover Type is a crucial parameter in urban climate studies as it is essential for distinguishing between urban and built-up

Table 1
Madrid's meteorological stations used in this study with the identification of their latitude, longitude, altitude, period of retrieval, LCZ, and classification as an urban or rural station.

Station ID	Station	Latitude (°N)	Longitude (°E)	Altitude (m)	Period	LCZ	Station Type
230	Retiro	40.412	−3.678	667	2000–2022	LCZ 2/B	Urban
3839	Torrejon	40.483	−3.450	611		LCZ 8/D	Rural
3946	Barajas	40.467	−3.556	609		LCZ 8/D	
3947	Cuatro Vientos	40.378	−3.789	687		LCZ 5/D	
3948	Getafe	40.300	−3.723	617		LCZ 6/D	

surfaces and other natural or rural surfaces to assess the impact of urbanisation on atmospheric variables. In this study, the selected Land Cover Type product (MCD12Q1) was obtained from the combined Moderate Resolution Imaging Spectroradiometer (MODIS) aboard the Terra and Aqua satellites. MODIS provides high radiometric sensitivity (12 bit) in 36 spectral bands ranging in wavelength from 0.4 μm to 14.4 μm with a nominal resolution of 250 m at nadir, with five bands at 500 m and the remaining 29 bands at 1 km. The yearly Land Cover Type is available from 2001 to 2020 at a 500 m spatial resolution over a global scale (1200 km × 1200 km). The product provides 17 land cover classes, displayed in the legend of Fig. 1 (Sulla-Menashe and Friedl, 2018). This product has been shown to accurately portray the land cover (Hu and Brunsell, 2015; Hutengs and Vohland, 2016; Schwarz et al., 2011; Van Nguyen et al., 2015; Wu et al., 2019), including in the Mediterranean region (Bachantourian et al., 2022). For this study, the land cover product for the year 2015 was specifically chosen to align with the spatial development of Madrid within the urbanisation context.

2.4. Land surface temperature

The analysis of Land Surface Temperature (LST) and the Surface Urban Heat Island (SUHI) effect is conducted using the LST product obtained from the Satellite Application Facility on Land Surface Analysis (LSA–SAF; Schmetz et al., 2002). The LST was obtained from the outgoing thermal infrared radiation (TIR) spectral band (8–13 μm), measured at the top of the atmosphere by the Spinning Enhanced Visible and Infrared Imager (SEVIRI) on board the Meteosat Second Generation (MSG) satellites. This dataset is available only under clear-sky conditions, in which the TIR spectral band is appropriate due to its weak atmospheric attenuation and the inclusion of the Earth's spectral radiance peak (Ermida et al., 2019).

The LST algorithm employed is based on the generalised split-window formulation, which was initially developed for the Advanced Very High-Resolution Radiometer and MODIS, and further adapted to SEVIRI split-window channels. According to this method, LST errors depend on the uncertainty of the surface emissivity, the atmospheric water vapour content, and the satellite view angle (Freitas et al., 2010). The LSA-SAF LST product is available at ~3 km resolution at the nadir, every 15-min, spanning from 2004 to the present day. The temperature values retrieved for the MSG disk are in the range of −80 °C to +70 °C (Schmetz et al., 2002).

The LSA-SAF LST product was selected considering its high temporal resolution (15 min), which allows to describe the daily cycle of LST and SUHI in Madrid at the hourly scale". Its lower spatial resolution (~5 km over Europe) in comparison to other LST products retrieved from satellites such as Landsat, ASTER, MODIS, and Sentinel, still allows the analysis of city-scale phenomena like the SUHI effect. The application of the LSA-SAF LST product for studying SUHI in previous studies include, for example, Nogueira et al. (2022) to characterise the Paris SUHI, and Zakšek and Oštir (2012) to downscale LST to analyse the SUHI diurnal cycle in central Europe. Moreover, the LSA-SAF LST quality has been extensively evaluated against in-situ stations data (Göttsche et al., 2016; Niclòs et al., 2011; Sun and Pinker, 2007; Trigo et al., 2008) showing good performance.

Firstly, the retrieved LST dataset was restricted to the study area (square in Fig. 1), a 20 × 20 matrix of 0.05° resolution (4.09°E –

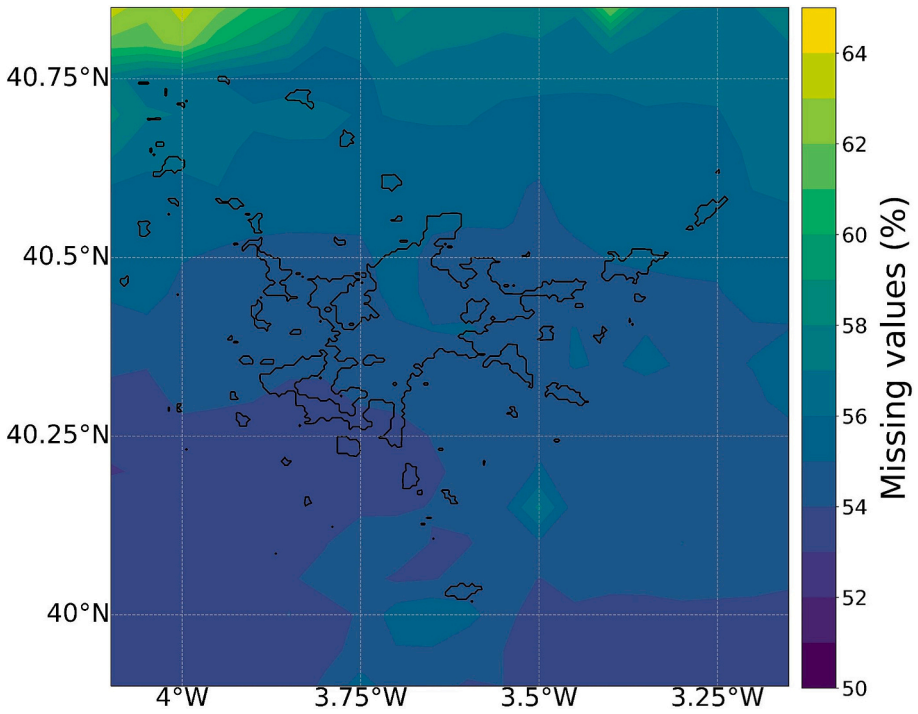


Fig. 2. Map of missing LST values average over the study area for the 2004–2022 period. The black contour delimits the Madrid's "Urban and Built-up" land cover from the MCD12Q1 product retrieved from MODIS for 2015.

3.15°E; 39.9°N – 40.85°N) for the 19-year period of available data (2004–2022). Secondly, the LST hourly mean was calculated by averaging the 0-, 15-, 30-, and 45-min estimates within each hour. Thirdly, the obtained dataset was treated by removing the lower and higher outliers calculated by: $Q1 - (1.5 \times IQR)$ and $Q3 + (1.5 \times IQR)$, respectively. In these expressions, Q1 and Q3 represent the first and third quartiles, respectively; IQR stands for interquartile range (difference between Q3 and Q1). The resulting hourly LST shows missing values ranging from 53% to 64% over the study area (Fig. 2) in the 2004–2022 period. Over the "Urban and Built-up" land cover type of Madrid and near surroundings (obtained from the MCD12Q1 MODIS product for 2015, black contour in Fig. 2), the minimum percentage of missing values is 54%, meaning that less than half of the total data is available for this study. These limitations are a common issue in studies that use thermal LST products since they are only available for clear-sky conditions. Therefore, the characterisation of LST and SUHI effects is constrained to clear-sky conditions and does not capture their full variability. Still, cloudless days in Madrid are mostly observed in summer (Núñez-Peiró et al., 2021), when both LST and SUHI are more intense (Michel et al., 2021; Núñez Peiró et al., 2017; Núñez-Peiró et al., 2021; Rasilla et al., 2019). It is worth noting that, in this study, XGBoost was trained with treated datasets by removing the missing values.

Afterwards, for each LST grid point (0.05°), a group of 12 × 12 points from the Land Cover dataset (0.005°) was considered to classify the LST points as rural, mixed, or urban. Then, the percentage of "Urban and Built-up" land cover was determined among each group of 12 × 12 points and attributed to the corresponding LST point. Following Nogueira et al. (2022), each LST point was classified as urban if the percentage of "Urban and Built-up" was above 66%, mixed for percentages within 33% and 66%, and rural if below 33%.

Fig. 3 shows the urban, rural, and mixed grid LST points represented as black, orange, and white circles, respectively; the contouring colours represent the urban fraction. The yellow star represents the urban centre of mass calculated from the contour line delimiting the urban area (>66%). Furthermore, the urban cluster was defined as a circle centred in the urban centre of mass with a radius slightly overpassing the farthest urban LST point (smaller circle in Fig. 3). The outer circle delimits a ring with the same area as the inner circle with the aim to delimit the rural area surrounding Madrid for maintaining a certain balance in the number of urban and observations. Take into consideration that the continuum of the MODIS LULC has been applied to the urban/rural classification. Within the defined limits by the circles aiming to separate urban and rural in equal areas, only the points with an urban fraction >66% were considered urban (the remaining mixed and rural points in the defined urban cluster by the circle were not contemplated). The same approach was applied to classify rural LST points, in which the mixed points were not contemplated within the defined rural belt. Therefore, this this approach allows to carefully separate the urban and the surrounding rural LST points. With this classification, the LST grid points were divided into 13 urban, 16 mixed, and 371 rural (46 of which are within the outer ring) points. In this study, the evaluation of the XGBoost LST was performed encompassing all rural points (371), whereas for the SUHI effect, the difference between the mean of 13 urban and the mean of 46 rural points of the outer ring was considered.

Noteworthy, when compared to the use of urban continuum of the city and the surrounding municipalities, the use of the representation system of urban and rural areas in this study is considered adequate for assessing the Madrid's climate, since cities have a

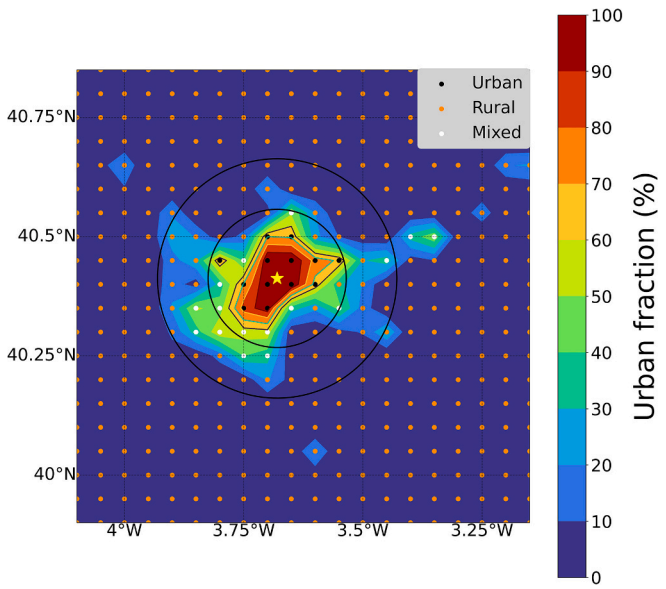


Fig. 3. Percentage of urban fraction (Urban and Built-up land) of MODIS land cover type within each LST grid box (LSA SAF). The classified urban, mixed and rural LST are represented by black, white, and orange points, respectively. The yellow star represents the calculated urban centre of mass. For SUHI computation, the inner circle delimits the urban area constrained by the urban fraction criteria (it is centred on the urban centre of mass and has a radius that slightly overpasses the farthest urban LST point). The outer circle delimits a ring with the same area as the inner circle, e for characterising the rural area. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

high surface heterogeneity pattern (Oke et al., 2017) which are not contemplated in the former municipal borders.

Moreover, following Nogueira et al. (2022), the 11:00 to 18:00 UTC and the 00:00 to 7:00 UTC periods were considered to evaluate the daytime and nighttime LST, respectively. For each month of the 2004–2022 period, a diurnal cycle counting was computed to ensure that the observed daily maximum and minimum LST occurs within the predefined intervals. The frequency of occurrence of the maximum and minimum LST at a given hour is depicted in Fig. 4a and Fig. 4b, respectively, showing the largest bars within the predefined intervals (shaded areas), confirming Nogueira's assumption. The equivalent figures for ERA5 and XGBoost are presented in Supplementary Material as Fig. S1, showing similar results.

2.5. ERA5 reanalysis

ERA5 is the latest generation (5th) of global climate reanalysis developed by the European Centre for Medium Range Weather Forecasts (ECMWF) as part of the Copernicus Climate Change Service (Hersbach et al., 2020). Covering the period from 1940 onwards, ERA5 provides a high-resolution dataset of the global atmosphere, land surface, and ocean-wave, with temporal, horizontal, and vertical resolutions of 1 h, ~31 km, and 137 levels extending from surface to 0.01 hPa, respectively. Using the ECMWF Integrated Forecast System Cy41r2, ERA5 benefits from enhanced model parametrisations and the four-dimensional variational data assimilation scheme compared to its predecessor, ERA-Interim (Dee et al., 2011). Outstandingly, ERA5 demonstrates improved accuracy in several climate variables, including near-surface air temperature, LST, wind, and radiation (Johannsen et al., 2019; Nogueira et al., 2020; Urraca et al., 2018). Additional details are available in Hersbach et al. (2020).

The analysis of Madrid's present climate ERA-5 features and its use as predictors for XGBoost is performed considering the 2000 to 2022 period at 0.25° horizontal resolution (regridded subset of the full ERA5 dataset at native resolution). Prior to training XGBoost with ERA5 predictors, the nearest neighbour between each observational dataset (meteorological station or LST) and the ERA5 data was determined. The closest ERA5 grid point was attributed to the corresponding observation location and applied for training.

In addition to the predictors (presented in Section 2.8) used in the XGBoost training, the hourly maximum and minimum 2-m temperatures and the hourly skin temperature (temperature of the uppermost surface layer, responding instantaneously to surface flux changes due to its lack of heat capacity) were retrieved from ERA5 to assess the added value of the XGBoost predictions. To address elevation differences between the local ground stations and the closest ERA5 grid-point to each station, an adiabatic temperature adjustment was applied to the ERA5 grid-points. A constant lapse rate of 6.5 K/km was assumed for this correction in agreement with many studies (Cardoso and Soares, 2022; Careto et al., 2022; Prein et al., 2015; Rhoades et al., 2020; Soares et al., 2022). Additionally, this correction was also performed for each of the meteorological stations, converging the altitude to the Retiro's station altitude.

2.6. Surface and urban heat islands computation

The urban heat island effect is defined as the temperature difference between the urban and nearby rural sites. This phenomenon is typically computed for 2-m air temperatures (UHI) and for land surface temperatures (SUHI). In this study, Madrid's UHI was defined as the temperature difference between the urban station (Retiro) and each of the rural meteorological stations considered (Torrejon, Barajas, Cuatro Vientos, and Getafe). The limited number of installed meteorological stations in cities (including Madrid) is a known shortcoming for urban studies, which hinders the depiction of the complex urban spatial heterogeneity. This issue is partially overcome by computing the SUHI, as LST is retrieved at a high spatial resolution. Here, SUHI was defined as the land surface temperature difference between the considered average urban (13 points) and rural (46 points of the outer ring) areas (Fig. 3; see Section 2.4 for details).

2.7. Extreme gradient boosting (XGBoost)

The Extreme Gradient Boosting (XGBoost) is a supervised Machine Learning technique that implements the gradient-boosted

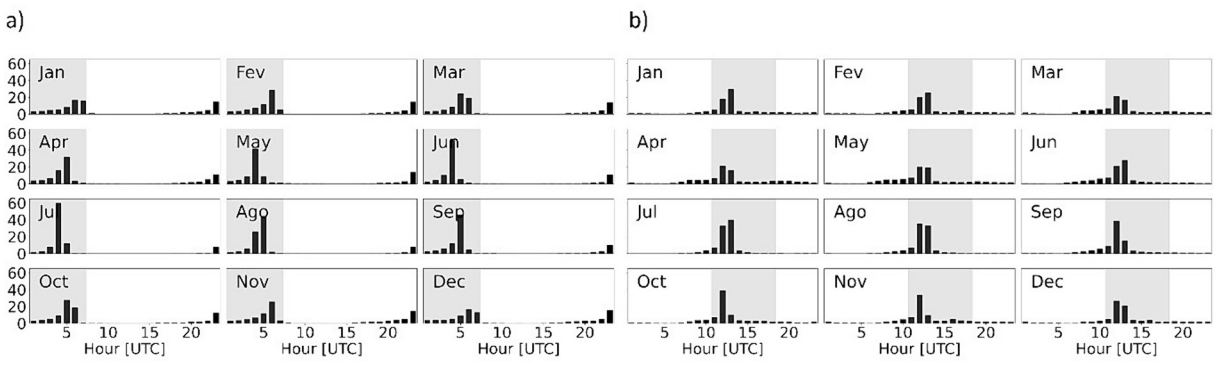


Fig. 4. Number of times that in a specific hour of the day corresponds to the (a) maximum and (b) minimum daytime observed LST presented for all months.

decision tree method, enabling the capture of non-linear relationships between predictors and observations therefore creating predictions from training data (Chen and Guestrin, 2016). This method can be applied to regression, classification, ranking, and other user-defined predictions. XGBoost has the advantage of avoiding overfitting and achieves optimal calculation speed through parallel computing during the training process.

The boosting method consists of building a strong predictive model through a series of iteratively generated weaker models. Firstly, an initial model is built using the training data. Then, the second model is built, aiming to correct the residual errors of the initial model. The process repeats, adding the generated models until it reaches a performance threshold or the predefined maximum number of models (Chen and Guestrin, 2016). The XGBoost algorithm implemented in this study is briefly explained by the following steps:

1) To implement XGBoost, the training and testing datasets are defined assuming 2/3 and 1/3 ratios of the total dataset, respectively. Additionally, a differentiable loss function is selected to evaluate the model's performance:

$$\begin{aligned} &\text{Data} && D = \{(x_i, y_i)\}_{i=1}^{n} \\ &\text{Loss function} && L(y_i, \gamma) = \frac{1}{n}\sum_{i=1}^{n}(y_i - \gamma)^n \end{aligned} \tag{1}$$

where x_i and y_i refer to the set of predictors and predictands, respectively, and i assumes values from 1 to n (which refers to the length of the dataset). The predictions are represented by γ. Typically, for regression applications and as performed in this study, the Mean Squared Error (MSE) is the considered loss function. The XGBoost was implemented considering the default parameters. The three key hyperparameters were assumed as: 100 for the number of gradient boosted trees; 6 for the maximum tree depth for base learners; and 0.3 for the boosting learning rate;

2) The model is initialised with a constant value (F_0) determined by the (γ) that minimizes the loss function $L(y_i, \gamma)$:

$$F_0(x) = argmin_\gamma \left(\sum_{i=1}^{n} L(y_i, \gamma) \right) \tag{2}$$

3) The residual of the model $(r_{i,m})$ is computed as the derivative of the loss function with respect to the predicted value:

$$r_{i,m} = -\left[\frac{\partial L(y_i, F(x_i))}{\partial F(x_i)}\right]_{F(x)=F_{m-1}(x)} \quad r_{i,m} = -(y_i - \gamma_i) = y_i - \gamma_i, \text{for } i = 1, \ldots, n \tag{3}$$

A regression tree is built, predicting the residuals of the first model. The terminal elements of each m tree are denominated "leaves" $(R_{j,m})$, where the index j refers to the leaf number. Given that each leaf can have more than one residual value, it is necessary to determine the final output value for each leaf $(\gamma_{j,m})$, calculated as the γ that minimizes the summation:

$$\gamma_{j,m} = argmin_\gamma \sum_{x_i \in R_{i,j}} L(y_i, F_{m-1}(x_i) + \gamma) \tag{4}$$

4) The last step is the update of the model prediction formed by the initial prediction F_0 and the "weighted" contribution of the calculated $\gamma_{j,m}$ residual outputs (multiplied by the selected learning rate):

$$F_m(x) = F_{m-1}(x) + \nu \sum_{j=1}^{J_m} \gamma_m I(x \in R_{jm}) \tag{5}$$

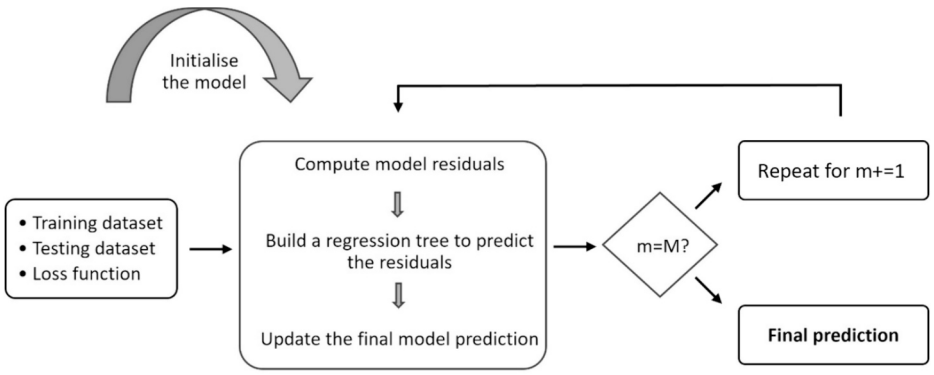


Fig. 5. Flowchart of the XGBoost architecture. The "m" and "M" represent, respectively, the number of tree iteration and the maximum number of trees (defined 100 as default).

Here, ν is the learning rate, assumed as 0.3 in this study.

5) Then, the third to fifth steps are repeated until the number of trees reaches the selected maximum number of trees (M = 100). Fig. 5 displays a flowchart that summarises the described XGBoost methodology.

2.8. Selection of predictors, training and evaluating

The selection of predictors in this study followed the rationale outlined in Zhao et al. (2021) and Zheng et al. (2021) studies and others urban investigations as Nogueira et al. (2020), Nogueira et al. (2022). The former authors incorporated atmospheric forcing fields that influence urban climate, including the net shortwave radiation (SW), net longwave radiation (LW), and total precipitation at the surface (TP); and the atmospheric temperature, pressure, specific humidity, and wind speed (zonal and meridional) at the forcing height. The full set of predictors considered in this study is listed in Table 2. This set of predictors follows, in some extent, the relevant energy flows that modulate the UHI effect through the SW and LW variables (Phelan et al., 2015).

Six different combinations of ERA5 predictors were built to perform sensitivity tests, to assess the dependency of XGBoost results on the set of predictors. The first set incorporated only surface-level predictors (SW, LW, and TP). Subsequently, the combination of surface level predictors with variables (Q, Z, T, U, and V) at different heights (40 m, 80 m) and pressure levels (1000 hPa, 925 h Pa, and 850 hPa) generated five additional predictors sets. These new sets were created to assess the added value of a more complete dataset and to evaluate the XGBoost's sensitivity to altitude-level meteorological information. The choice of 40 m and 80 m heights was based on Nogueira et al. (2020) and Nogueira et al. (2022) studies, where these heights were used to assess the UHI effects in Paris and Lisbon, respectively, in a land surface offline modelling approach. The 1000 hPa and 850 hPa pressure levels were selected based on their general availability among ESGCM. These tests enable an assessment of XGBoost's sensitivity to different pressure level predictors for a further application to future climate studies. Finally, the 925 hPa level was considered since it closely matches Madrid's urban canopy height. Additionally, the XGBoost was trained with monthly, daily, and hourly (only for LST) timescales producing predictions of Madrid's present climate variables for equivalent timescales, allowing a better understanding of the sensitivity of XGBoost to different temporal scales. As aforementioned, to describe the present urban climate, the predictors were selected at a 0.25° spatial resolution. Additionally, encompassing the application to future climate studies, the XGBoost was trained with the same selected datasets at 1° resolution, given the present ESGCM available resolution. With this approach the sensitivity of XGBoost to different spatial resolutions was also assessed.

The dataset was divided into the training set (2/3 of the total dataset) and the testing set (the remaining 1/3), according to the temporal range of available data for each predictand (Table 3). In the first part of the study, the XGBoost's ability in predicting the present climate of Madrid was evaluated by training the model with the ERA5 variables (predictors) and the observed variables: T_{min}, T_{max} and LST (predictands). Subsequently, the trained XGBoost model was applied to predict daily T_{min}, T_{max} and hourly LST in the testing periods.

The XGBoost ability in simulating local climate over Madrid was evaluated for the testing period by computing four error metrics: Mean Bias (Bias); Mean Absolute Error (MAE); Standard deviation ratio; and Perkins skill score (Sscore) which provides the similarity score between the modelled and observed probability density functions (Perkins et al., 2007). For this aim, an extensive analysis of the different XGBoost setups (trained with a diverse set of predictors and at different ERA5 resolutions) was performed for daily T_{min} and T_{max} and hourly LST. The results of XGBoost downscaled predictands from ERA5 were compared with the corresponding observational datasets and the ERA5 reanalysis. Since the XGBoost downscaling used ERA5 predictors, its evaluation was performed with daily synchronized data. Finally, annual cycles, including a seasonal view of all the urban features was evaluated, as well as mean diurnal cycles of LST and SUHI. The seasonal scales are focused on the typical summer and winter seasons (DJF – December, January, and February; JJA – June, July, and August) and on the usually designated extended summer and winter, which include all months (NDJFMA – November, December, January, February, March, and April; MJJASO – May, June, July, August, September, and October). The use as well of extended seasons (results presented in the Supplementary Material) is justified by the amplification of the seasonal length detected in temperature as a result of global warming (Christidis et al., 2013; Pena-Ortiz et al., 2013; Santer et al., 2018).

Table 2
ERA5 predictors with the considered temporal period, height and pressure levels, time scales and spatial horizontal resolution.

Variables		Period	Level	Time Scale	Spatial Resolution
Net Surface solar radiation	SW	2000 to 2022	Surface	Hourly Daily Monthly	0.25° 1°
Net Surface thermal radiation	LW				
Total precipitation	TP				
Specific Humidity	Q		80 m		
Geopotential/Atmospheric Pressure	Z/P		40 m		
Temperature	T		1000 hPa		
U-component of wind	U		925 hPa		
V-component of wind	V		850 hPa		

Table 3
Total, training and testing periods of predictors dataset for present climate predictands (2-m air temperature and land surface temperature).

Predictand	Total period	Training period	Testing period	Time scale
2 m-air temperature	[2000,2022] 23 years	[2000, 2015] 16 years	[2016, 2022] 7 years	Monthly Daily
Land surface temperature	[2004, 2022] 19 years	[2004, 2016] 13 years	[2017, 2022] 6 years	Monthly Daily Hourly

3. Results

3.1. Predictors sensitivity cases: evaluation of daily temperatures and hourly LST

In this section, an evaluation of different XGBoost setups is performed for daily T_{min} and T_{max} and hourly LST.XGBoost is forced by ERA5 predictors at 0.25° and 1° spatial resolutions, using surface level and at height/pressure levels data. Fig.6 displays the error metrics for Tmax at the yearly scale and for winter and summer seasons, computed for the testing period (2016–2022) across all meteorological stations. The configurations including surface-level predictors with 40 m and 80 m are presented at a 0.25° resolution,

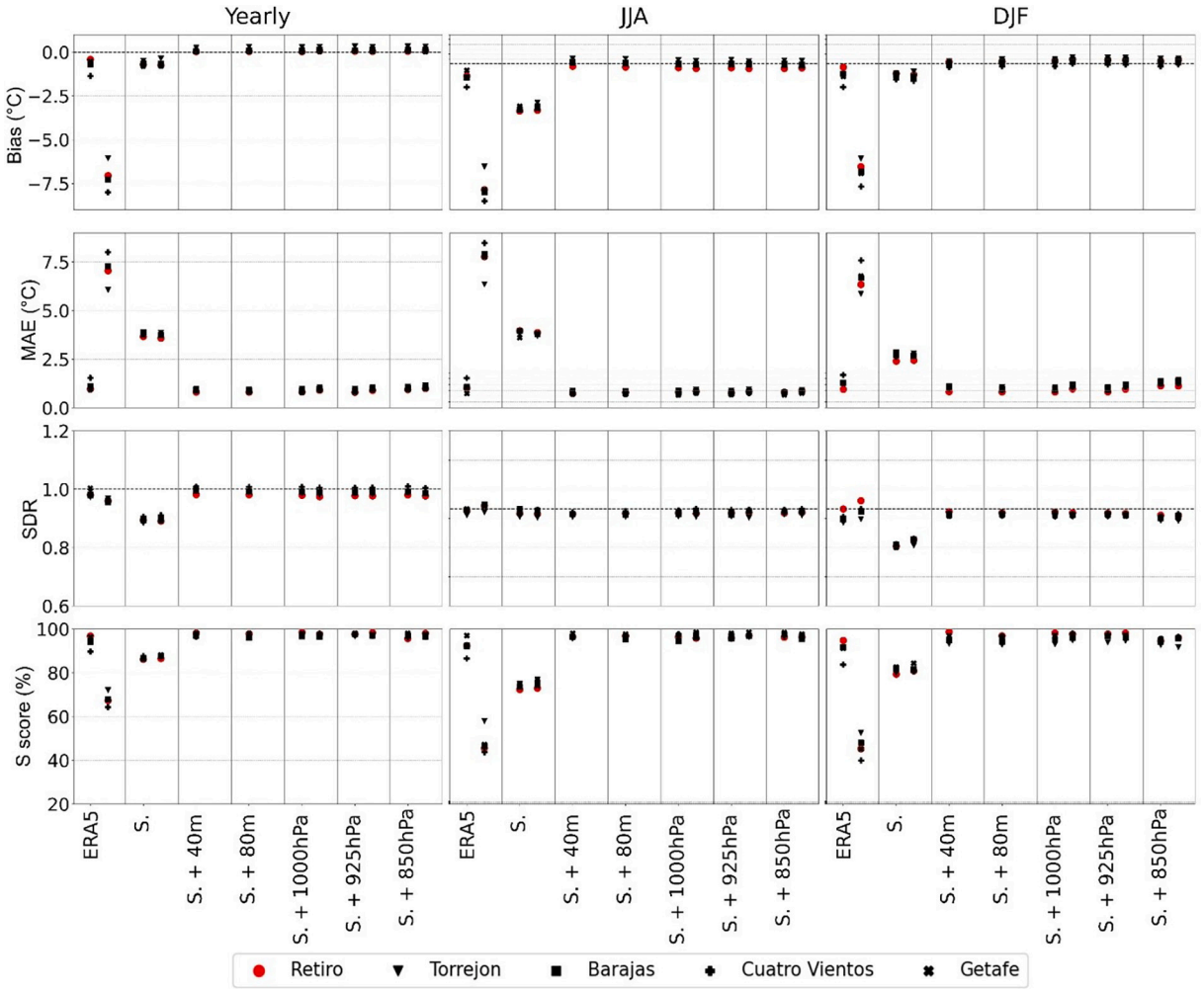


Fig. 6. Daily maximum 2-m daily temperature errors metrics computed over the test period (2016–2022) for ERA5 and XGBoost (Surface, Surface+40 m, Surface+80 m, Surface+1000 hPa, Surface+925 hPa, and Surface+850 hPa) sensitivity cases. The sensitivity cases are presented at 0.25° (left on each column) and 1° (right on each column), except for Surface+40 m and Surface+80 m, that are shown only for 0.25°. The error metrics (a) Bias, (b) MAE, (c) Standard deviation ratio and (d) Sscore are presented for the urban (Retiro) and rural (Torrejon, Barajas, Cuatro Vientos and Getafe) meteorological stations of Madrid. From top to bottom, the rows represent the errors computed for the Year, Winter, and Summer seasonal scales (DJF and JJA).

while the other sensitivity tests are shown for both 0.25° (on the left) and 1° resolutions (on the right).

The first two columns depict the results for ERA5 and XGBoost, trained only with surface-level predictors, respectively, exhibiting overall suboptimal outcomes for both spatial resolutions. In contrast, notably improved performance is observed when considering the error metrics for XGBoost trained with surface-level predictors combined with data from other vertical levels (40 m, 80 m, 1000 hPa, 925 hPa, and 850 hPa). For these cases, the biases are around 0 °C for all stations. MAE presents values of about 1 °C for the yearly scale for both seasons. The normalised standard deviation error is approximately 1.0 for the yearly mean, while for summer and winter seasons, the values range between 0.9 and 1.0. Regarding the S score, error metrics approach 100% for the annual scale, but for individual seasons, the results exhibit large scores above 90%. Furthermore, no individual station stands out in the results, as all demonstrate similar performances.

Additionally, when comparing error metrics computed at 0.25° and 1°, all XGBoost sensitivity cases present similar results, even when using only surface-level predictors. For example, considering MAE for the annual scale, ERA5 presents an error increase of over 6 °C on average between values at 0.25° and 1° (< 2 °C and 8 °C, respectively). The rather small sensitivity of XGBoost results to predictors resolution, whether using height- or pressure-level data, offers confidence in its use with different input data and for future climate applications.

Similar findings are found for T_{min}, as illustrate in Fig.7. ERA5 stands out once more as the worst product. XGBoost trained only with surface-level predictors exhibits significant disparities compared to the combined sensitivity cases with other vertical levels, as

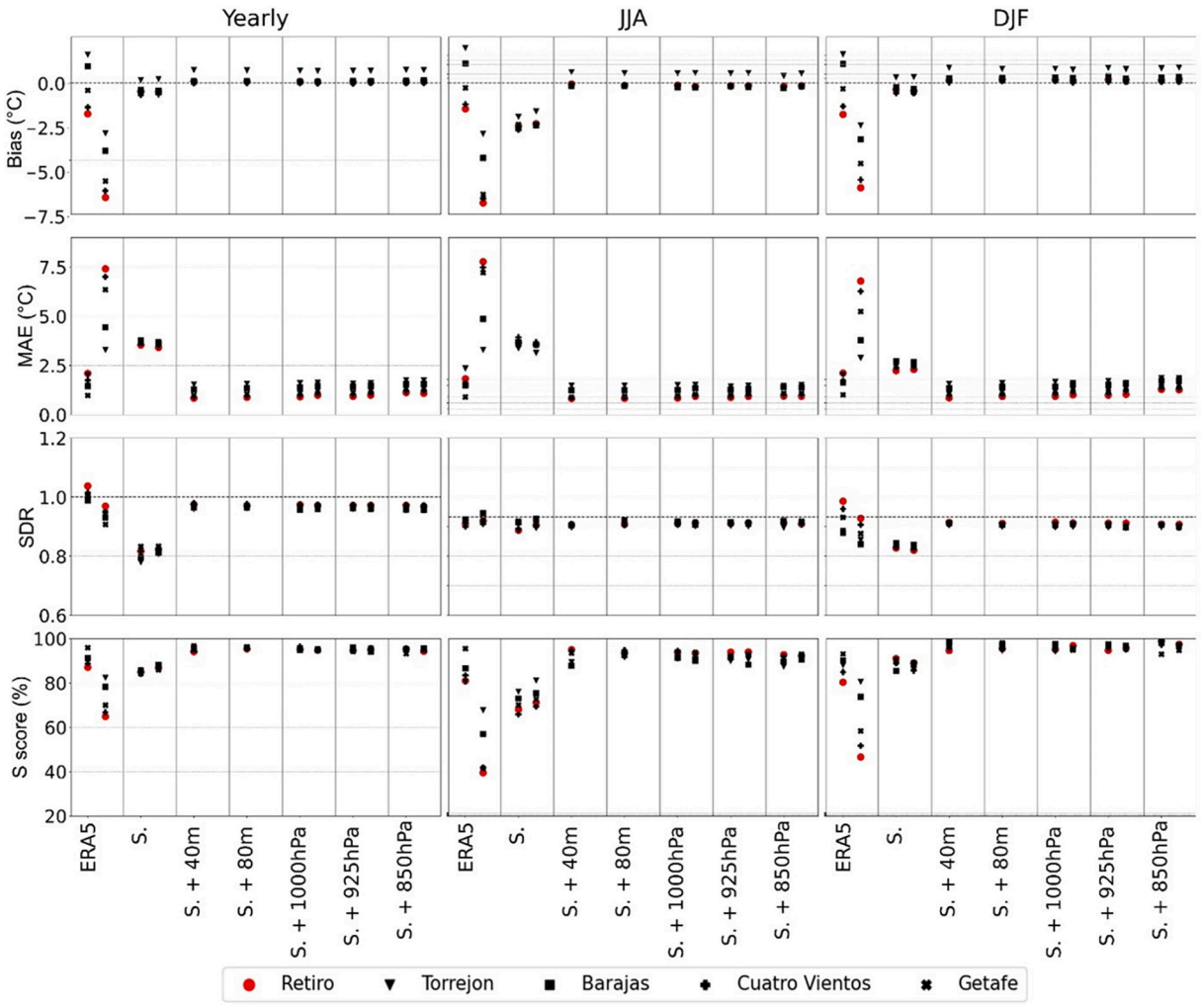


Fig. 7. Daily minimum 2-m daily temperature errors metrics computed over the test period (2016–2022) for ERA5 and XGBoost (Surface, Surface+40 m, Surface+80 m, Surface+1000 hPa, Surface+925 hPa, and Surface+850 hPa) sensitivity cases. The sensitivity cases are presented at 0.25° (left onof each column) and 1° (right onof each column), except for Surface+40 m and Surface+80 m, that are shown only for 0.25°. The error metrics (a) Bias, (b) MAE, (c) Standard deviation ratio and (d) Sscore are presented for the urban (Retiro) and rural (Torrejon, Barajas, Cuatro Vientos and Getafe) meteorological stations of Madrid. From top to bottom, the rows represent the errors computed for the Year, Winter, and Summer seasonal scales (DJF and JJA).

observed for T_{max}. This suggests that while the employed surface variables (SW, LW, and TP) play a crucial rule in characterising physical urban processes, incorporating vertical levels predictors in the training dataset enhances greatly XGBoost performance. This enhancement stems from XGBoost's capacity to learn non-linear relationships during training.

Concerning the presented results, summary tables were produced to identify the most accurate predictor configuration for XGBoost, considering seasonal (Summer, Winter, Extended Summer, and Extended Winter) and yearly values for each error metric. Table 4 depict the mean error metrics computed for the selected meteorological stations at 0.25° for T_{max} and T_{min}. In each table, the best XGBoost case is highlighted for each season and metric (blue shading). The cumulative depiction of the best cases is provided in the last column. By analysing both tables, it is evident that the combined set of surface predictors with 40 m-level predictors stands out as the most accurate XGBoost configuration for predicting both target variables.

A similar assessment was conducted for LST computed over the testing period (2017–2022) at 0.25° resolution. Fig. 8 illustrates the hourly LST errors considering the full year, Summer and Winter, divided into rural, mixed, and urban grid points. In general, both ERA5 and XGBoost trained only with surface-level predictors reveal clear shortcomings, as seen for the 2-m maximum and minimum temperature. In fact, the surface-level XGBoost presents, overall, less accurate results compared to ERA5 (e.g., MAE medians of +7.5 °C for all land covers in JJA vs ERA5's + 4 °C values). Nonetheless, XGBoost sensitivity cases with vertical levels information present remarkable results with relatively low variability. The Bias medians are around +1 °C for DJF and −1 °C for JJA, while the MAE values are approximately 2 °C for all cases. All land cover types exhibit standard deviation ratio values close to 1, indicating that the observed standard deviation of LST is well represented by XGBoost. Furthermore, 95% of the observed PDF is represented by XGBoost considering the yearly LST, which slightly reduces to 90% for winter and summer seasons.

Overall, there is no noticeable difference among the combined XGBoost sensitivity cases across the analysed error metrics (Table 5).

Table 4

Summary for mean (a) 2-m maximum daily temperature (b) 2-m minimum daily temperature errors computed over the test period (2016–2022) for the XGBoost (first column: Surface-Surface, 40 m-Surface+40 m, 80 m-Surface+80 m, 1000 hPa-Surface+1000 hPa, 925 hPa-Surface+925 hPa, and 850 hPa-Surface+850 hPa) sensitivity cases at 0.25° resolution. The blue shading represents the best sensitivity case for each season (Yearly, DJF, JJA, Extended Summer and Extended Winter) and error metric (Bias, MAE, Standard deviation ratio and S score). The total number of best cases is shown in the last column.

(a)

	Bias					MAE					Standard deviation ratio					S score					
	Year	DJF	JJA	Summer	Winter	Year	DJF	JJA	Summer	Winter	Year	DJF	JJA	Summer	Winter	Year	DJF	JJA	Summer	Winter	Total
Surface	-0.686	-0.738	-2.766	-2.614	-1.468	3.802	2.676	3.864	4.153	3.552	0.895	0.618	0.956	1.042	0.787	86.669	76.429	67.485	79.584	56.523	0
40m	0.088	-0.002	0.107	0.144	0.083	0.895	1.007	0.777	0.805	0.938	0.995	0.943	0.942	0.97	0.996	97.385	94.907	95.965	97.146	68.769	6
80m	0.108	0.084	0.072	0.134	0.145	0.86	0.945	0.782	0.785	0.882	0.993	0.942	0.948	0.971	0.996	97.266	94.286	95.662	96.683	68.856	1
1000 hPa	0.097	0.097	0.018	0.103	0.139	0.859	0.937	0.777	0.791	0.885	0.992	0.943	0.954	0.971	0.995	97.498	94.814	95.418	96.989	69.738	3
925 hPa	0.128	0.188	0.017	0.111	0.194	0.859	0.945	0.766	0.786	0.879	0.989	0.939	0.955	0.971	0.992	97.414	95.807	95.911	96.865	70.112	4
850 hPa	0.128	0.115	-0.02	0.094	0.17	0.993	1.27	0.762	0.831	1.136	0.992	0.911	0.968	0.983	0.998	96.608	93.075	96.318	96.45	69.265	6

(b)

	Bias					MAE					Standard deviation ratio					S score					
	Year	DJF	JJA	Summer	Winter	Year	DJF	JJA	Summer	Winter	Year	DJF	JJA	Summer	Winter	Year	DJF	JJA	Summer	Winter	Total
Surface	-0.427	-0.298	-2.723	-2.476	-1.144	3.667	2.483	3.643	3.955	3.389	0.808	0.705	0.901	0.993	0.685	84.916	86.315	63.292	79.304	65.377	2
40m	0.235	0.342	0.057	0.129	0.256	1.119	1.166	1.062	1.083	1.117	0.971	0.932	0.916	0.957	0.963	95.55	95.443	90.093	94.301	75.358	8
80m	0.231	0.363	0.004	0.109	0.28	1.17	1.227	1.076	1.123	1.182	0.969	0.919	0.933	0.963	0.964	95.756	95.159	91.46	94.627	75.521	4
1000 hPa	0.216	0.379	-0.026	0.073	0.256	1.203	1.261	1.103	1.166	1.213	0.967	0.92	0.935	0.965	0.96	95.315	94.498	90.994	94.41	75.468	3
925 hPa	0.216	0.389	-0.035	0.079	0.27	1.223	1.328	1.104	1.159	1.247	0.968	0.914	0.933	0.962	0.962	94.973	94.966	89.938	93.649	75.489	0
850 hPa	0.24	0.35	-0.131	0.082	0.276	1.401	1.567	1.187	1.308	1.476	0.966	0.915	0.935	0.959	0.97	94.961	96.444	88.292	93.183	75.698	3

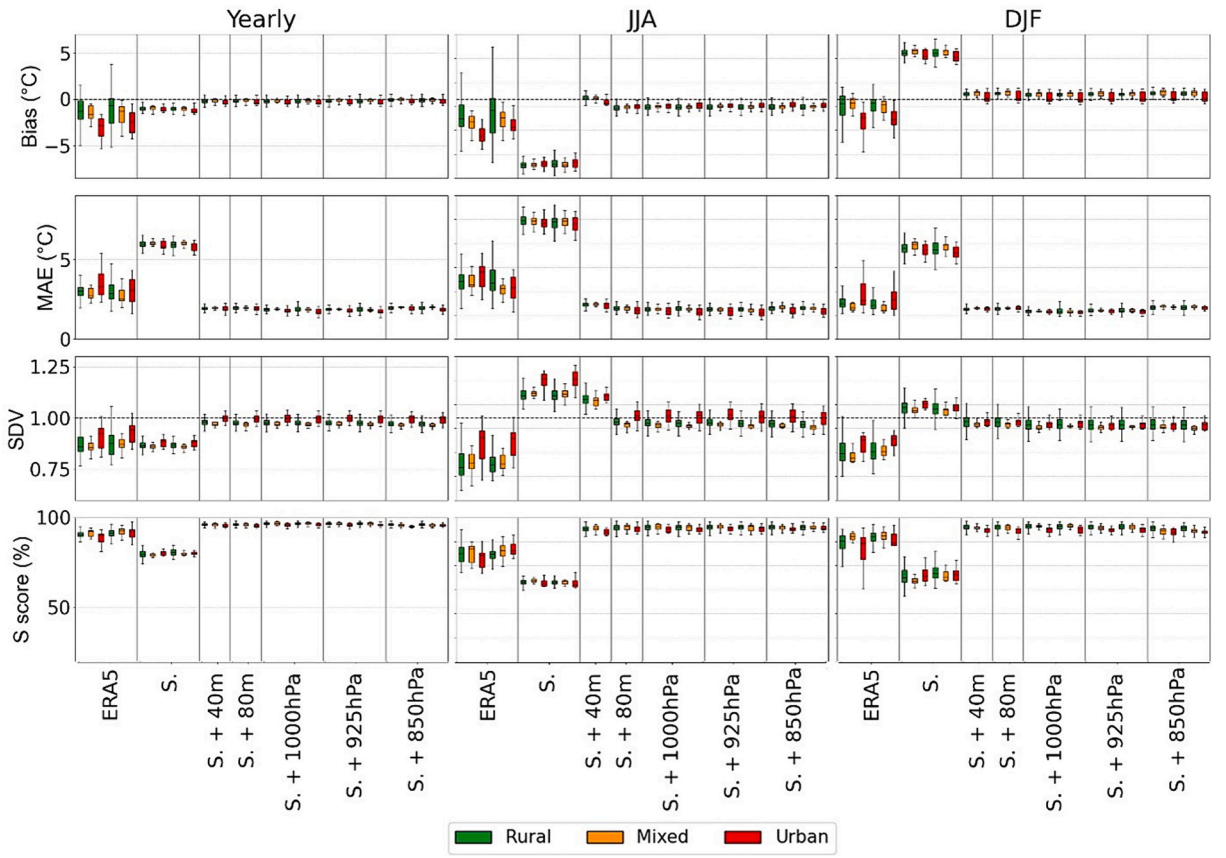


Fig. 8. Boxplots of Land Surface Temperature errors computed over the testing period (2017–2022), at hourly scale, for ERA5 and XGBoost (Surface, Surface+40 m, Surface+80 m Surface+1000 hPa, Surface+925 hPa and Surface+850 hPa) sensitivity cases. The latter are presented at 0.25° (left onf each column) and 1°(right onf each column), except for Surface+40 m and Surface+80 m, that are shown only for 0.25°. The boxplots represent the (a) Bias, (b) MAE, (c) Standard deviation ratio and (d) S score for all grid boxes classified as rural surfaces (grid boxes with urban fraction below 0.33), mixed surfaces (grid boxes with urban fraction between 0.33 and 0.66), and urban surfaces (grid boxes with urban fractions above 0.66), represented as green, orange, and red, respectively. From top left to bottomright, the rows columns represent the errors computed for the Yearly, Winter and Summer seasonal scales (DJF and JJA). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 5
Summary for mean rural, mixed, and urban LST errors computed over the test period (2017–2022) for the XGBoost (first column: Surface-Surface, 40 m-Surface+40 m, 80 m-Surface+80 m, 1000 hPa-Surface+1000 hPa, 925 hPa-Surface+925 hPa, and 850 hPa-Surface+850 hPa) sensitivity cases at 0.25° resolution. The blue shading represents the best sensitivity case for each season (Yearly, DJF, JJA, Extended Summer and Extended Winter) and error metric (Bias, MAE, Standard deviation ratio and S score). The total number of best cases is shown in the last column.

	Bias					MAE					Standard deviation ratio					S score					
	Year	DJF	JJA	Summer	Winter	Year	DJF	JJA	Summer	Winter	Year	DJF	JJA	Summer	Winter	Year	DJF	JJA	Summer	Winter	Total
Surface	-1.009	4.983	2.924	-4.783	4.573	5.942	5.717	5.403	6.148	5.638	0.871	1.053	1.085	1.037	1.106	79.838	67.092	83.516	74.941	75.235	0
40m	-0.213	0.493	-0.027	-0.501	0.215	1.925	1.905	2.144	1.914	1.942	0.985	0.98	1.095	0.996	1.015	95.903	94.005	93.307	95.777	94.643	3
80m	-0.183	0.55	0.024	-0.494	0.277	1.941	1.938	2.135	1.93	1.958	0.983	0.979	1.091	0.994	1.014	95.864	93.81	93.699	95.682	94.727	0
1000 hPa	-0.227	0.397	-0.009	-0.481	0.148	1.833	1.743	2.045	1.848	1.812	0.984	0.966	1.086	0.993	1.005	96.433	94.333	94.236	96.017	95.253	9
925 hPa	-0.195	0.457	-0.025	-0.446	0.176	1.837	1.796	2.044	1.829	1.849	0.984	0.964	1.086	0.994	1.002	96.28	93.99	94.212	96.096	94.925	4
850 hPa	-0.1	0.571	0.063	-0.37	0.3	1.948	2.028	2.101	1.907	2.008	0.979	0.966	1.082	0.986	1.002	95.73	92.631	94.273	95.858	94.236	4

Equivalent plots performed for the Extended Summer and Winter seasons show similar results (not shown). Additionally, the LST analysis is also performed at a daily scale (not for all hours), encompassing both daily maximum and minimum LST (derived from the hourly values). The corresponding results are similar for both daily maximum and minimum LST, however, overall, slightly inferior to the hourly scale results (Bias of -2 °C and $+$ 2 °C for JJA and DJF, respectively and MAE medians of $+2$ °C).

Summarizing, in the case of T_{max} and T_{min}, XGBoost trained with the combined surface and 40 m-level predictors set present the best results with a total of 14 best cases out of 40. For LST, the surface+1000 hPa predictor set is the best-performing one despite the rather small differences observed among the various cases (Table 5). In studies focusing on dynamical urban climate, the height level immediately above the urban canopy is commonly chosen as the height of the forcing data. Hence, in this study, the combined surface and 40 m-level predictors set is chosen to train XGBoost and characterise the Madrid's present climate, as detailed in the following subsection.

3.2. Spatial and temporal characterisation of T_{min}, T_{max} and LST

In this section, the 2-m temperatures and LST are characterised in terms of temporal and spatial scales. Firstly, the average annual cycles of T_{max} (Fig. 9a, b) and T_{min} (Fig. 9c, d) were computed for Retiro and Torrejon stations for the testing period (2016–2022). The curves illustrate the expected annual temperature cycles, with warming from January to July followed by cooling from August to December. A large discrepancy is observed between ERA5 and observations, particularly pronounced for T_{min}, reaching 3 °C in summer months. ERA5 tends to underestimate the annual cycle for both local stations for T_{max} and in Retiro for T_{min}, while it overestimates the T_{min} annual cycle in the Torrejon station. In turn, the XGBoost predictions describe very closely the observed temperature annual cycles, exhibiting only a slight overestimation for T_{min} in the Torrejon station. To simplifying the analysis, considering the aim of this study to propose a ML method for improving the urban climate characterisation, only the Retiro-Torrejon description is fully described given that these stations have the strongest UHI signal among all combinations in this set of stations (see Fig. S1 in Supplementary Material). The results for the other considered rural stations (Barajas, Cuatro Vientos, and Getafe) are presented in Supplementary Material (Fig. S2).

Fig. 10 shows the annual cycle of the hourly LST (at monthly scale) for the training period (2017–2022); the data is clustered based on land cover type: rural, mixed, and urban. As expected, there are no major differences observed in the annual cycles of LST between the rural, mixed, and urban, except for the increasing intensity from the rural to urban cycles (with maximum values exceeding 30 °C

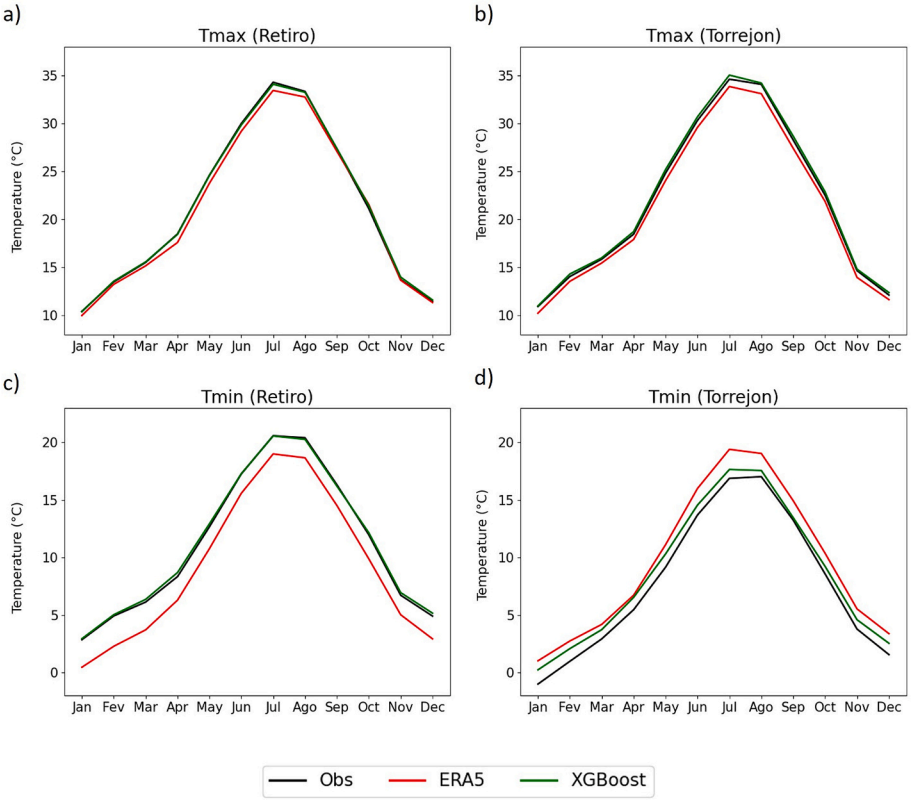


Fig. 9. Average annual cycle of observations (black line), ERA5 (red line) and XGBoost (green line) for (a) Tmax and (b) Tmin for Retiro (urban) and Torrejon (rural) stations. The results are presented for the testing period (2016–2022). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

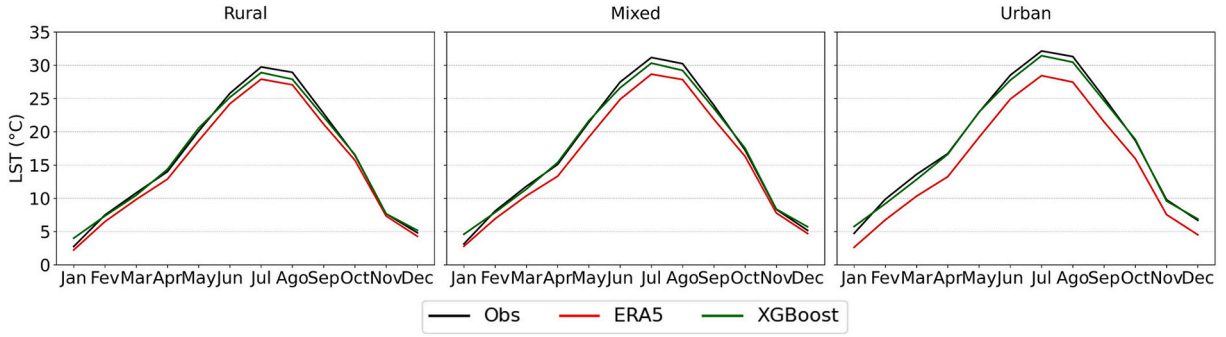


Fig. 10. Average annual cycle of observed (black line), ERA5 (red line) and XGBoost (green line) LST computed over the testing period (2017–2022) for (a) rural surfaces (urban fraction below 0.33), (b) mixed surfaces (urban fraction between 0.33 and 0.66), and (c) urban surfaces (urban fractions above 0.66). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

for urban and remaining below 30 °C for rural points). Both ERA5 and XGBoost underestimate the annual LST cycle in all cases. However, XGBoost performs much better than ERA5, especially for the urban case, where ERA5 displays a negative bias of approximately −5 °C.

Due to the hourly temporal resolution of LST, an analysis of the mean daily cycle is also performed (Fig. 11). The accuracy of XGBoost in describing the mean diurnal cycle is strikingly good, producing an almost identical diurnal cycle to the observed LST for all land cover types. However, XGBoost results present a slight shift in the timing of the maximum LST in all cases. In contrast, ERA5 shows a much smaller diurnal amplitude, underestimating the diurnal maximum LST by >5 °C, and as well a slight delay in surface heating/cooling after sunrise/sunset.

To spatially characterise LST, maps of mean daytime (11:00 to 18:00 UTC) and nighttime (00:00 to 7:00 UTC) LSTs computed over the test period (2017–2022) for the yearly, winter, and summer seasons are presented in Figs. 12 and 13, respectively. A clear similarity is noticed between the observed and XGBoost-predicted spatial patterns of daytime and nightime LST. A clear observational SUHI pattern is evident over Madrid, more pronounced during DJF (with temperatures of about 17 °C in the urban core and between 14 °C and 16 °C in the rural surroundings) that is also captured in the XGBoost predictions. Moreover, for JJA, a weaker cool island is spotted over the centre and northeast of the city. For all seasons (spring and autumn not shown), XGBoost reproduces well the spatial patterns of LST and the subsequent SUHI effect. Furthermore, ERA5 fails completely to simulate a SUHI effect over Madrid, presenting a homogeneous and colder average temperature over the whole area of study. Regarding the mean nighttime observed LST, a strong signature of the SUHI effect is visible for the (Fig. 13). XGBoost demonstrates again notable accuracy in predicting the SUHI intensity and its spatial distribution, while ERA5 displays constantly colder (warmer) temperatures in the northern (southern) half of the city, revealing no signs of a SUHI effect.

3.3. Daily urban heat island effect

The full timeseries and boxplots for the training dataset (2016–2022) are computed for the maximum and minimum daily UHI between Retiro and Torrejon stations (Fig. 14), and the other rural stations. The left panels of Fig. 14 highlight the clear inability of ERA5 to represent the observed UHI pattern for both T_{max} and T_{min}. This behaviour is confirmed in the right panels of the Figure. For T_{max}, ERA5 UHI medians indicate a weak and positive UHI intensity (around +0.2 °C) while the observed and XGBoost UHI distribution is negative, with median values around −0.8 °C and reaching values below −2 °C. In the case of T_{min}, ERA5 shows a substantial

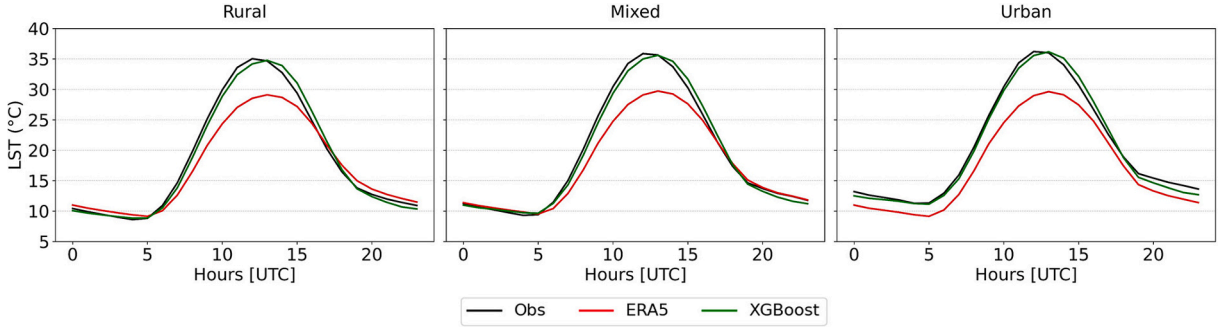


Fig. 11. Average diurnal cycle of observed (black line), ERA5 (red line) and XGBoost (green line) LST for (a) rural (urban fraction below 0.33), (b) mixed (urban fraction between 0.33 and 0.66), and (c) urban (urban fractions above 0.66) LST. The results are presented for the testing period (2017–2022). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

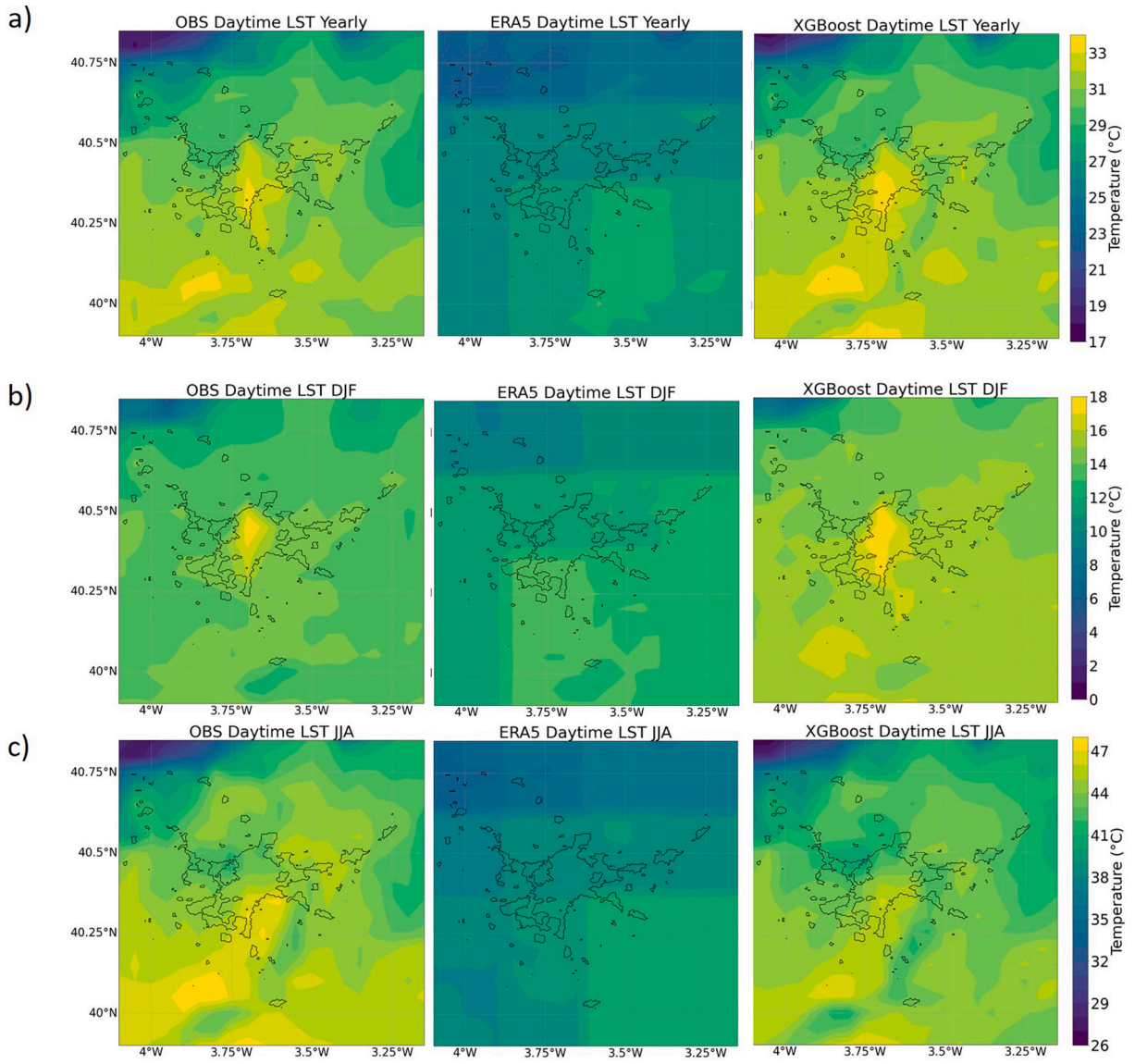


Fig. 12. Maps of observed (left), ERA5 (middle) and XGBoost (right) mean daytime (11:00 to 18:00 UTC) LST: (a) Yearly, (b) winter (DJF), and (c) summer (JJA).

underestimation of the UHI intensity, with ERA5 UHI values just above 0 °C, compared to the observed UHI median values around +2.8 °C. XGBoost performs rather well in simulating the observed UHI, including the median values and the full variability. However, some discrepancies are noted in the timeseries, particularly in representing some extreme observed values. The highest UHI intensities are observed for Torrejon (+2.5 °C for T_{min} UHI), followed by Barajas (+2.5 °C for T_{min} UHI, not shown) stations. The remain UHI presents medians of T_{min} UHI intensities below +1 °C.

The seasonal daytime and nighttime UHI are also computed for the observations, ERA5 and XGBoost predictions (Fig. 15) considering all urban-rural differences. ERA5 fails to simulate both daytime and nighttime UHI for all seasons. Similarly to the previously analysed yearly scale results (Fig. 14), for daytime, the observed seasonal UHI has a relatively weak and negative signal, with values ranging from −0.6 °C to −0.3 °C throughout the year. Conversely, the observed nighttime UHI intensity is positive and constant throughout the year (+1.5 °C), except for DJF, where the UHI is slightly stronger, reaching values above +2 °C. XGBoost displays a striking performance in predicting the UHI effect, closely following the observed UHI signal and intensity for all seasons, both daytime and nighttime. However, a small underestimation is noticeable for the daytime JJA UHI.

3.4. Daily and hourly surface urban heat island effect

In this sub-section, the analysis of the SUHI effect is displayed. The SUHI mean values (considering the 46 rural and 13 urban grid

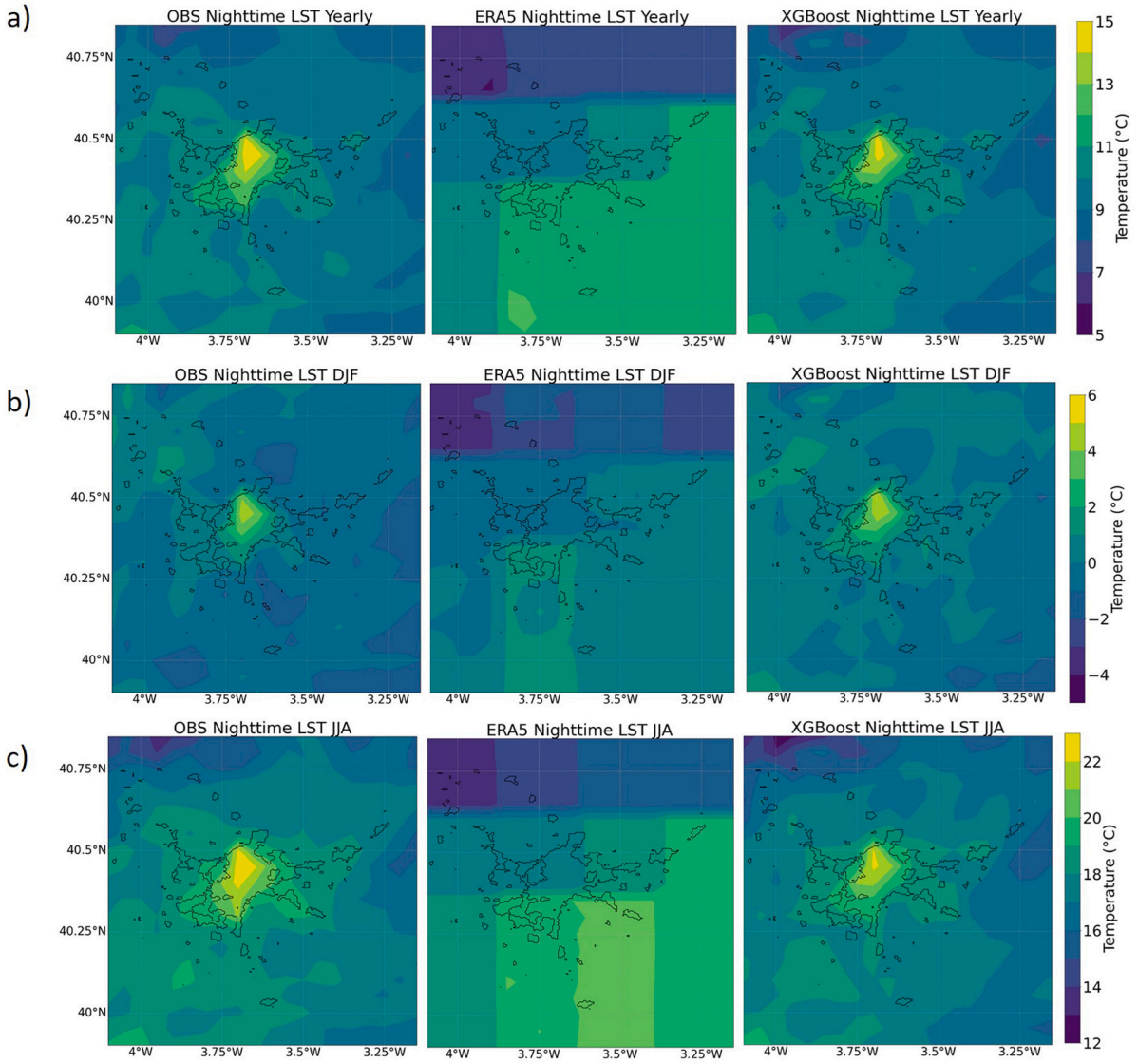


Fig. 13. Maps of observed (left), ERA5 (middle) and XGBoost (right) mean nighttime (00:00 to 07:00 UTC) LST: (a) Yearly, (b) winter (DJF), and (c) summer (JJA).

points) are computed for the determined maximum and minimum LST from the hourly dataset considering the 11:00 to 18:00 UTC and 00:00 to 07:00 UTC periods (Fig. 16). The results are revealed for each season and for the yearly scale. The maximum daytime SUHI exhibits a seasonal pattern characterised by a stronger intensity in spring with values around +1.5 °C. The other seasons display a relatively constant SUHI intensity, slightly above and below +1 °C for DJF and SON, respectively. An exception is observed for JJA, where the daytime maximum SUHI is slightly negative (−0.1 °C). For the nighttime SUHI, the intensity remains relatively constant throughout the year with values around +2 °C, peaking in JJA. Overall, the XGBoost demonstrates a great performance in simulating the SUHI seasonal values, significantly outperforming ERA5. However, the XGBoost daytime SUHI in summer reveals an overestimation of ∼0.5 °C and a reverse signal (positive SUHI) – an inaccuracy not present in other seasons. In this case, ERA5 shows a good performance, although this behaviour is explained by error compensation, since ERA5 displays constant SUHI values around 0 °C throughout the year. For the nighttime minimum SUHI, XGBoost has a better overall performance, slightly underestimating the observed effect in all seasons.

The mean diurnal cycle of SUHI was computed for all seasons (Fig. 17). In agreement with Núñez-Peiró et al. (2021), immediately after sunrise (which occurs in Madrid between 5:00 and 8:00 UTC, depending on time of the year), the rural area warms rapidly as it receives direct solar radiation. The urban area, in turn, warms at a slower rate, given its thermal proprieties and the influence of urban canyons that may obstruct direct solar radiation. This causes a rapid decrease of the SUHI intensity starting from sunrise. Notice that this effect is stronger during summer, explained by the stronger solar radiation input. The enhanced heating capacity of the adjacent

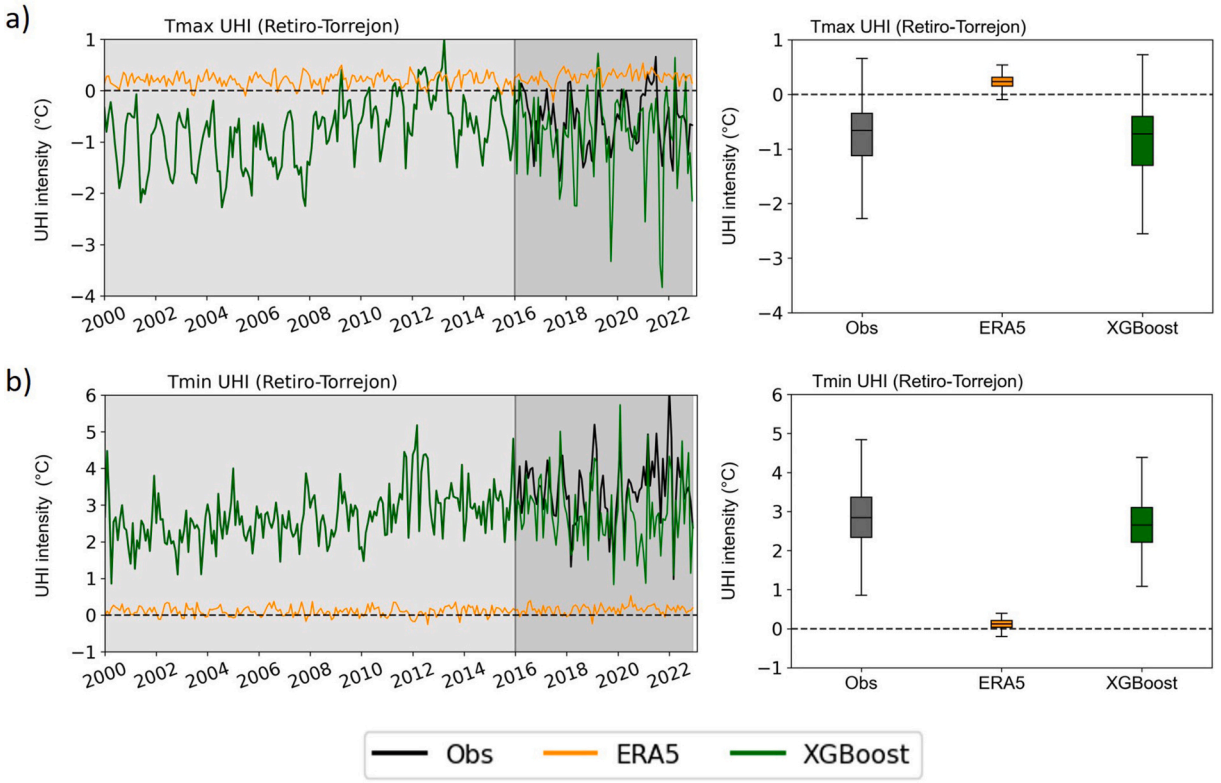


Fig. 14. Timeseries and boxplots computed for observed, ERA5 and XGBoost at daily scale (a) maximum and (b) minimum UHI (presented in the left and right panels, respectively), calculated between Retiro and Torrejon stations. The boxplot results are presented for the testing period (2016–2022).

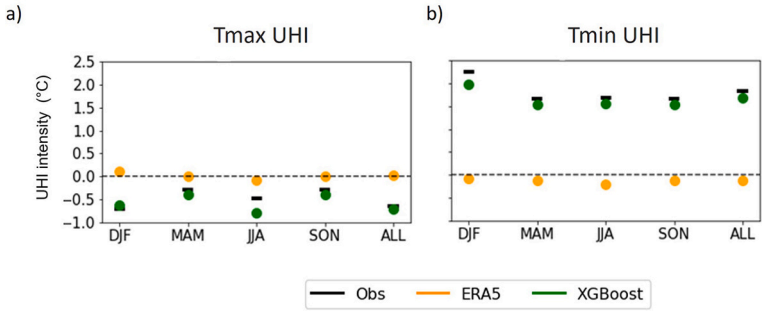


Fig. 15. Seasonal (DJF, MAM, JJA, SON) and ALL-Yearly observed, ERA5 and XGBoost (represented as black lines and orange and green scatters, respectively) UHI total averages computed for (a) daily maximum and (b) daily minimum temperatures. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

arid rural landscape creates a negative SUHI effect in summer (reaching −0.5 °C). Later in the morning, around 10:00 UTC, the urban area becomes warmer than the rural surroundings, inverting the SUHI variation rate. Then, by midday, the urban and rural areas reach their temperature peak, decreasing the heating rate and, consequently, slowly increasing SUHI. From midday to afternoon, both surfaces start to cool and given their physical characteristics, the urban surface lags behind the rural area. SUHI, therefore, continues to slowly increase in intensity. ERA5 shows a clear incapacity to represent the diurnal cycle of SUHI, with a nearly constant behaviour and values between 0 °C and − 0.5 °C for every season. In opposite manner, XGBoost improves greatly the diurnal cycle representation. However, XGBoost exhibits a slight general underestimation of the observed effect during the night and late afternoon, along with an overestimation during the morning and early afternoon. The SUHI overestimation is larger for JJA, by >1 °C between ~9:00 and ~ 15:00 UTC, thus failing to simulate the characteristic signal inversion (cool island) of Madrid's SUHI during summer.

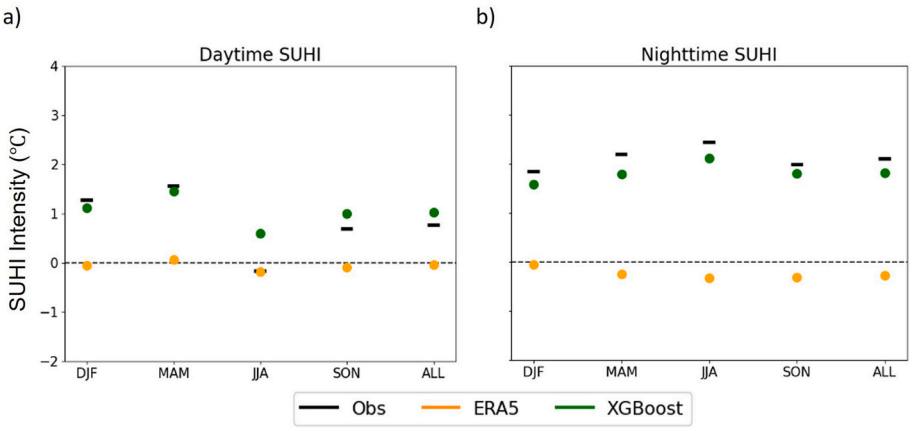


Fig. 16. Seasonal (DJF, MAM, JJA, SON) and ALL-Yearly observed, ERA5 and XGBoost (represented as black lines and orange and green scatters, respectively) SUHI total average computed for (a) daytime (11:00 to 18:00 UTC) and (b) nighttime (00:00 to 07:00 UTC). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

4. Discussion

In this study, the potential advantage of applying XGBoost (a machine learning algorithm) was explored to describe relevant features of urban climate, focusing on 2-m air temperature, surface temperature and the subsequent UHI and SUHI effects. The largest city of Iberia – Madrid – was selected as the study case. XGBoost was applied to describe the urban temperature-related variables of the present climate of Madrid. The algorithm was trained using a set of ERA5 predictors and the target observations, encompassing data from local ground stations for the 2000–2022 period and remote sensing data for the 2004–2022 period. The training and testing datasets were split into 2/3 and 1/3 ratios, respectively.

The XGBoost predictions for Madrid's present climate variables were produced at daily and hourly (only for LST) time scales, but also at monthly time scale (not shown). This temporal analysis aimed to enhance our understanding of XGBoost's sensitivity across different temporal scales. In fact, for all time scales, the method presented remarkable consistency and accuracy in simulating Tmax, Tmin and LST. Given the considerable temperature variations in Madrid, encompassing both daily and yearly amplitudes, the robust performance of XGBoost suggests its potential applicability for describing climate features to a plethora of other cities under distinct climatic backgrounds. This poses an additional advantage in the context of the lack of hourly and daily long-term observational datasets for some locations.

Given the innovative application of XGBoost for urban settings and to describe urban climate properties, a number of sensitivity cases focused on the selection of the predictors for XGBoost was performed. The results of the sensitivity tests underscored the limited efficacy of XGBoost when trained only with surface-level predictors. This suggests that while the surface-level fluxes are physically relevant variables in the context of urban climate, these variables alone are insufficient for an accurate XGBoost performance. In fact, the results obtained with surface-level predictors combined with other pressure levels predictors are outstandingly better for all combinations. Furthermore, these combinations reveal rather good results for the evaluated error metrics, representing a clear improvement over ERA5, particularly for Tmax, Tmin and LST at the different annual and seasonal scales. The sensitivity of XGBoost to different spatial resolution of predictors was also tested with the aim to understand its potential application to train with coarser resolution (1°) ESMGCMs data for climate change analysis. The XGBoost trained with the full set of predictors presented similar results for both spatial resolutions (0.25° and 1°). Thus, the low sensitivity of the method to different pressure levels and spatial resolutions of predictors, validates its applicability across a multitude of scenarios, such as ESGCMs with different available height-pressure levels. However, it is important to note that the assessment of the present climate of Madrid was here shown for the best-performing setup, namely, the combined "surface + 40m" predictors set.

At daily and hourly temporal scales, XGBoost outperforms ERA5 in reproducing the yearly cycle of Tmax, Tmin and LST. Additionally, when representing the LST diurnal cycle, the XGBoost's performance was equally accurate for rural, mixed, and urban grid points, improving the ERA5 underestimation of LST (by over 5 °C) at the peak of the day. Similar results were also found at monthly scale. XGBoost, in close agreement with observations, revealed intensities of −0.5 °C and + 1.5 °C for daytime and nighttime UHI throughout the year (mean of all stations). These intensities are similar to the ones obtained in other studies. A study on the UHI behaviour for 10 dryland cities (Dialesandro et al., 2019) evidenced that Madrid and Dubai were the only ones demonstrating a strong and negative daytime UHI signal with urban temperatures >2 °C below rural. This effect might be explained by the rapid heating characteristics of surrounding arid, agricultural and bare soil areas, creating urban cold islands. Nevertheless, this effect is not restricted to semi-arid or desert climates, also occurring in regions with different climates, especially when surrounded by agricultural and bare soils. During nighttime, both cities display the strongest positive UHI effect among others (> +2 °C). The UHI variability over different cities is explained by its dependency on the degree of aridity, extent of vegetation, elevation, humidity, latitude, topography, and typical building types (Dialesandro et al., 2019). A recent study over Madrid (Núñez-Peiró et al., 2021) estimated the relationships

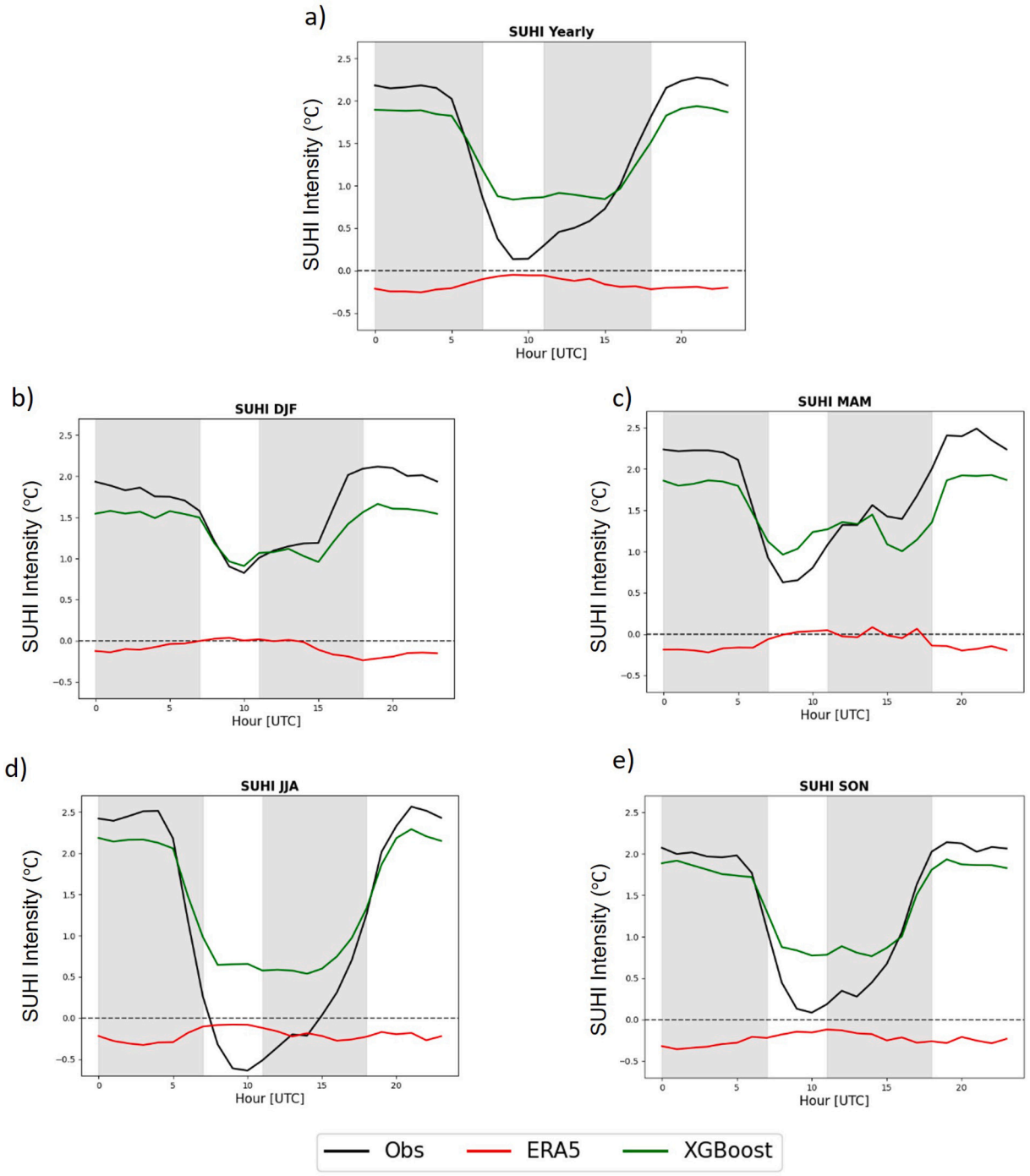


Fig. 17. Daily cycle of the observed (black line), ERA5 (red line) and XGBoost (green line) SUHI for the (a) yearly scale, (b) DJF, (c) MAM, (d) JJA and (e) SON seasons. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

between LCZ and UHI, evaluating their hourly and seasonal evolutions. Based on a monitoring campaign carried out over the period 2016–2018, the results showed that, as expected, temperature oscillations were greater during summer than during winter months. Moreover, during daytime, the temperature differences between most LCZs were small, becoming more visible during nighttime. Considering the daily cycle, temperatures started to differ greatly immediately after sunset (between 17:00 and 20:00 UTC), reaching their maximum disparity moments before sunrise (between 5:00 and 8:00 UTC).

In the current study, Torrejon and Barajas are the rural stations which presented the strongest urban-rural temperature difference with Retiro (for both daytime and nighttime). This behaviour is in agreement with Rasilla et al. (2019), which estimated the summer

season UHI and SUHI effects for Madrid and analysed the impacts of heatwaves on them for the 1980–2017 period. The authors explained that the farthest Torrejon (to the city centre) displays larger Tmin and Tmax UHIs, with +3 °C and − 1 °C intensities, respectively. Their UHI estimations for the Barajas station (surrounded by airport facilities) are slightly weaker for Tmin (~ +2 °C), however, a slightly greater intensity is revealed for Tmax (−1.75 °C). However, in the current study, the Tmax UHI effect exhibited similar values for both stations, around −0.5 °C; and, for Tmin, Torrejon presented a slightly greater intensity (+3 °C) compared to Barajas (+2.5 °C). These findings were obtained from the yearly mean computed over the period spanning 2016–2022, ie a different time period. The daily maximum and minimum UHI effects were not reproduced by ERA5 (null values throughout the year).

XGBoost's ability to describe Madrid's daily SUHI effect is remarkable. XGBoost represents a strong and positive nighttime SUHI intensity (around +2 °C), following the slight intensity variations of observations throughout the year. For daytime, XGBoost represented accurately the observed SUHI above 1 °C in winter and spring and below 1 °C in autumn. However, it fails to capture the little negative (−0.1 °C) Tmax SUHI in summer, exhibiting a bias of +0.5 °C. This behaviour is a matter of future investigation. A possible approach to overcome this issue might be the implementation of a seasonality parameter as performed in Zheng et al. (2021). Another avenue is to explore a wider range of predictors, such as latent and sensible heat flux, that might add value to the description of that feature. As highlighted by Rasilla et al. (2019), the spatial distribution of summer LST in Madrid generates a slight cold urban heart island effect (negative SUHI) with more emphasis over northern Madrid. This phenomenon is simulated by XGBoost with colder (~ − 1 °C) temperatures for urban LST. The results also showed that, overall, ERA5 does not properly reproduce the monthly and daily Tmax, Tmin, UHI and SUHI effects. This is mainly explained by the lack of representation of urban processes in the ECMWF Integrated Forecast System used to produce ERA5. Moreover, its relatively coarse grid resolution (0.25°) is not suitable for representing the highly heterogeneous urban land surface (LST and SUHI).

Finally, it is fair to recognize that further investigation is needed, namely, to determine the optimal predictor selection and configuration, to determine the optimal ratios for both the training and testing datasets and acknowledge that in the case of LST would be interesting to consider more than clear-sky values.

5. Conclusions

In this paper, an innovative AI method (XGBoost) is extensively explored for downscaling, for the first time, ERA5 and describe the Madrid's urban climate, foresee its applicability at a range on timescales, and characterise the most relevant urban climate properties, such as Tmax, Tmin, LST and the urban heat island and surface urban heat island effects. Departing from a set of ERA5 predictors, XGBoost is remarkably successful to describe all those urban features, specifically, daily scale Tmax and Tmin, and the related UHI, and hurly LST and the associated SUHI. Moreover, XGBoost shows striking results, w.r.t. the forcing ERA5 and observations, in reproducing from mean annual to daily cycles of the urban heat island effects.

Considering the XGBoost promising results, its application could be extended to other cities under different background climates (inland and coastal cities, mountainous areas or valley systems, polar cities, etc). A further investigation on the predictor's physical-based selection is essential to help to understand in a improved way how the XGBoost method works and assess its potential sensitivity to other factors. This insight could contribute to the development of a more physically robust training set, crucial for generating accurate projections for different cities. Additionally, future research directions could include optimizing XGBoost with specific hyperparameters, using bayesian optimization with cross validation, to potentially enhance its efficacy in generating urban climate predictions. This approach aims to find the optimal combination of three key hyperparameters: the number of gradient boosted trees, the maximum tree depth for base learners, and the boosting learning rate.

Finally, there is a clear lack of studies on the future evolution of Madrid's climate, under the different future emission scenarios. Moreover, a denser observation network is crucial for a better urban climate characterisation. The need for a better monitoring system of the weather in cities is urged by this study. The limited knowledge about air and surface temperatures, as well as the associated UHI and SUHI effects in Madrid, and its high vulnerability, demands an improved quantification of the city's climate properties in the context of a changing climate in order to develop mitigation and adaptation strategies to temperature extremes and heatwaves.

Funding

The publication of this research is funded by the Portuguese Fundação para a Ciência e a Tecnologia (FCT) I.P./MCTES through national funds (PIDDAC) – UIDB/50019/2020 – Instituto Dom Luiz.

CRediT authorship contribution statement

Angelina Bushenkova: Writing – original draft, Software, Methodology, Formal analysis. **Pedro M.M. Soares:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Frederico Johannsen:** Writing – review & editing, Methodology. **Daniela C.A. Lima:** Writing – review & editing, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

The authors would like to acknowledge the financial support of Portuguese Fundação para a Ciência e a Tecnologia (FCT) I.P./MCTES through national funds (PIDDAC) – UIDB/50019/2020 (https://doi.org/10.54499/UIDB/50019/2020), UIDP/50019/2020 (https://doi.org/10.54499/UIDP/50019/2020) and LA/P/0068/2020 (https://doi.org/10.54499/LA/P/0068/2020). Angelina Bushenkova was supported by a grant through the project "ELABORAÇÃO DA ESTRATÉGIA REGIONAL DE ADAPTAÇÃO ÀS ALTERAÇÕES CLIMÁTICAS DO ALENTEJO – ERAACA". FJ and DCAL acknowledge FCT I.P./MCTES (Fundação para a Ciência e a Tecnologia) for the FCT https://doi.org/10.54499/UI/BD/151498/2021.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.uclim.2024.101982.

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